

Quanta of Space

The geometric works of DICTO, in plain words

DICTO

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All works, findings and names are DICTO's. Computation, checking and drafting with Claude (Anthropic).

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The book covers 44 works: 42 papers in four chapters, the Kaleidohedra findings and the data set *DICTO Solids: Exact Geometry*. This draft holds the contents, the introduction, the whole of Chapter 1 (Atoms, 12 works), the Bonds entry #51 and their appendix sections. Each entry tells the work in plain words; every number, exact value and check is in the appendix at the back.

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In this draft: the introduction, all of Chapter 1 (Atoms) and #51. Order and chapters as in DICTO: Collected Geometric Works, Version 24 (DICTO DOI: 10.5281/zenodo.23345699).

INTRODUCTION

Quanta of Space



Is space smooth all the way down, like water? Or is it built from pieces, the way a wall is built from stones? People have asked this for two and a half thousand years. This book takes the question literally. It looks for exact sets of solid pieces that fit together with no gaps, into shells, into cycles and into the whole of space.

What follows is a family tree of an idea, not a claim about physics. The pieces in this book are shapes you could build and hold. They share a question with the old thinkers and with modern science: what are the smallest pieces of space, and how do they join?

Atoms and the empty space between them

In ancient Greece, Leucippus and Democritus taught that the world is made of tiny pieces that cannot be cut, the atoms, moving through empty space. The atoms differ only in their shape, their arrangement and their position. The Roman poet Lucretius put the teaching into verse: “nothing exists per se except atoms and the void”. So the atoms were pieces. But no one said what shapes they had.

Plato's kit of pieces

Plato gave them shapes. In his dialogue the *Timaeus*, the four elements are four regular solids. Fire is the tetrahedron, air the octahedron, water the icosahedron and earth the cube. All their faces are built from just two kinds of right-angled triangle. The fifth regular solid, the dodecahedron, was used “for the whole”. This was the first kit for building matter: a small set of exact pieces from which everything is put together.

Aristotle's mistake

Aristotle replied that if atoms are solid, they must fill space with no gaps. He claimed that only two solids do this: the pyramid and the cube. He was right about the cube. He was wrong about the regular tetrahedron, which does not fill space, and his error stood for centuries. Our paper #44, *Aristotle's Pyramid*, asks what he could have meant, and proves which pyramids cut from the regular solids can fill space.

Kepler's packings

In 1596 Johannes Kepler nested the five regular solids between the spheres of the planets. In 1611, in a small book on the six-cornered snowflake, he explained the snowflake by how equal small units pack together. That became the famous question of how tightly equal balls can be stacked. Our paper #8, *The Euclid–Kepler–Pacioli Cell*, finds all five regular solids nested on 13 points of one repeating cell.

The quanta of modern science

In 1900 Max Planck found that energy comes in small fixed amounts, quanta. In one modern theory of gravity, called loop quantum gravity, space itself does too. In 1995 Carlo Rovelli and Lee Smolin showed that in this theory area and volume come in separate steps, not smoothly. In 2011 Eugenio Bianchi, Pietro Doná and Simone Speziale showed that each small amount of volume can be read as a little solid with flat faces, fixed by the sizes and directions of its faces. Space becomes a network of such solids, joined face to face.

There is an honest difference. In that theory two neighbouring pieces share a face of the same area, but in general not of the same shape (Freidel and Speziale, 2010). Their pieces need not fit. Our pieces always fit exactly, corner to corner. Other approaches, such as causal sets, build space and time from points rather than from solids.

So this book makes no claim about physics. It shares the question and the form: what are the smallest exact pieces from which space can be built, and how do they join?

Four chapters

Read this way, the works fall into four chapters.

Atoms are single solids, each an exact new piece: the DICTO Ogdoecontahedron (OG), the Hecatodihedron, the Jewel, the Hexa, AXE, CLEO and FUJI, the Reuleaux Platonic solids, and the six DICTONIC SOLIDS (Icosaxes, Unborn Axes, Lotus Jewel Axes, Dodecaxes, Samson and Delilah). Their exact shapes are gathered in the data set *DICTO Solids: Exact Geometry*, where every DICTO solid has a permanent D-number.

Bonds are about how pieces fill space together: exact kits, packings and lattices, from the Euclid–Kepler–Pacioli Cell to the OG on every stacking. The clearest cases are the three DICTONIC BONDS (golden, keyed and Delian): pairs of solids that fill space together. A DICTONIC BOND is one such pair; the chapter Bonds holds every way pieces join.

Shells are pieces closing into nested layers round a core, like the skins of an onion, each layer filled exactly: the Chain of Karma, the Spheres of Dharma, the Bodhi Seed and Theaetetus.

Cycles are rules that turn one shell into the next and come back round, with or without end: Universal PiTet, the OG family steps, Kyushindo Cores and the shells turned inside out.

Between the repeating and the never-repeating

Some patterns repeat, like the tiles of a bathroom floor. Others never repeat, yet are perfectly ordered, like the Penrose tiling. The works in this book use the same pieces to build both.

For most of history, order in matter meant repetition. A crystal was one small cell copied without end, and five-fold symmetry was thought impossible in it. Then in 1982 Dan Shechtman found a metal alloy whose scattering pattern showed sharp five-fold symmetry. These quasicrystals, ordered but never repeating, earned him the Nobel Prize in 2011. Matter has two kinds of order, and the question became how they meet.

The pieces in this book sit on that border. One exact kit builds the repeating octet and also a crystal that never repeats, in the Fibonacci pattern (#40, *The DICTO OG Family on Every Stacking*). In *A Fibonacci Spine* (#30), five repeating packings of the OG meet round a five-fold axis. The copies that have to be taken out along that axis make exactly the Fibonacci chain, the simplest order that never repeats. The golden shells of the Spheres of Dharma (#23) repeat only by growing larger by the same golden factor at each turn. One kit, two kinds of order: the pieces themselves are the bridge.

What the pieces show

Three things come back again and again. A small kit is enough: a handful of exact shapes builds shells and spaces without limit. The golden ratio appears wherever five-fold symmetry does, and it decides when shells close. And the pieces join the two kinds of order in matter, the repeating and the never-repeating.

The ancients guessed that space was made of shaped pieces, and modern physics has come back to the idea. These works build such pieces exactly and test them, and each paper behind this book is checked exactly.

Sources for the introduction: Plato, *Timaeus* (tr. B. Jowett); Lucretius, *On the Nature of Things* (tr. W. E. Leonard); Aristotle, *On the Heavens* III.8 (tr. J. L. Stocks); J. Kepler, *Mysterium Cosmographicum* (1596) and *Strena seu de nive sexangula* (1611); D. Shechtman, I. Blech, D. Gratias, J. W. Cahn, Phys. Rev. Lett. 53 (1984) 1951; C. Rovelli, L. Smolin, Nucl. Phys. B 442 (1995) 593; L. Freidel, S. Speziale, Phys. Rev. D 82 (2010) 084040; E. Bianchi, P. Doná, S. Speziale, Phys. Rev. D 83 (2011) 044035; S. Surya, Living Rev. Relativ. 22 (2019) 5. Rewritten in plain words from the essay *Quanta of Space* in DICTO: Collected Geometric Works (DICTO DOI: 10.5281/zenodo.23345699). Licence CC BY-SA 4.0.

The DICTO Hexa

A cube made of eight Jewels, its partner the Hexa-Key, and a checkerboard that fills space

If you held it

You would hold a cube with 24 low roofs on its faces, pointing outwards. It looks, in the paper's words, “beautiful and solid, like carved out of wood”. Its surface shows 24 diamond-shaped windows, each the thick rhombus of Penrose's tiling.

It is made of eight DICTO Jewels, set on a two-by-two-by-two block and passing through each other inside. As one solid it holds as much as six and two-thirds Jewels.

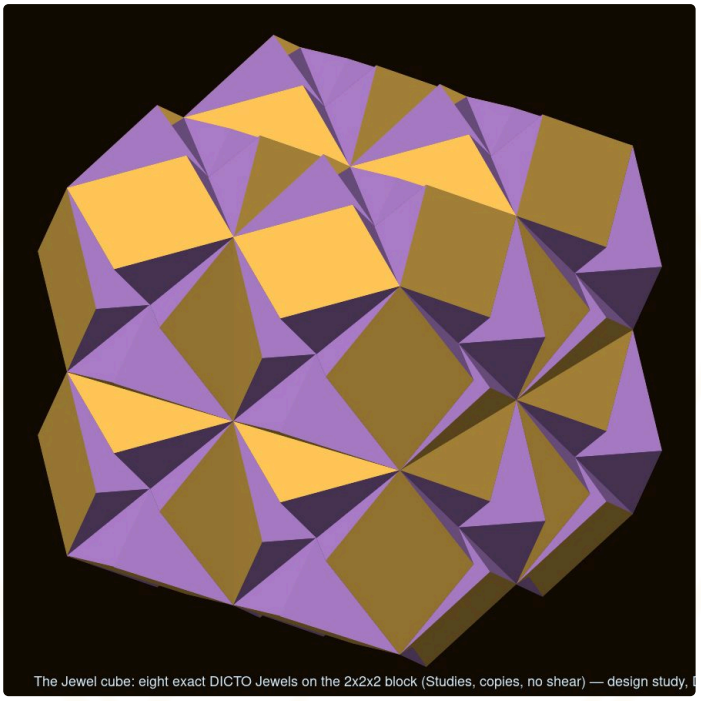


Figure 18. The DICTO Hexa: eight DICTO Hexa Jewels (exact DICTO Jewels on a 2 × 2 × 2 block), no shear. From the paper's Figure 1.

Where it started

The DICTO Jewel comes from the Euclid–Kepler–Pacioli Cell (#8). Take the cell's regular dodecahedron, which is a cube with Euclid's six roofs on it. Carve away the star-shaped stella octangulas of its six neighbouring cells. What remains keeps the whole cube and two-thirds of each roof, and its surface shows twelve Penrose rhombi, one on each edge of the cube: the windows. Jewels and stella octangulas, placed like the black and white cells of a checkerboard, fill space together.

We saw the cube of eight Jewels in the Studies world of the Kaleidohedra app, about eight Jewels in volume.

What we found

No amount of shearing makes eight separate Jewels meet face to face as a cube. The Hexa exists only as eight exact Jewels overlapping. Squashing the Jewels themselves loses the windows, so that was not kept.

Hexas meet each other only window to window, two whole windows per neighbour. In the face-centred cubic pattern each Hexa has 12 neighbours. The space left between them is one single solid, which DICTO named the **Hexa-Key**: eight stella octangulas and twelve double roofs. Hexas and Hexa-Keys then fill space like a checkerboard, five parts to three by volume, just as Jewels and stellas do one level down. There is a pleasing symmetry: the Hexa is eight cubes with 24 roofs turned out, and the Hexa-Key is eight stellas with the same 24 roofs turned in.

The Hexa also comes apart neatly, in two ways: into 8 cubes and 24 roofs, or into 4 whole Jewels and 4 trimmed ones. Eight Hexas can cluster round a sealed-in Hexa-Key, the Jewel rhombohedron of the Kagome Hulls one level up. In a diamond network, Hexas and stellas fill space with no single Jewels left over. And nothing turns: no join in these packings can be turned and still fit.

What it adds

The filling of space by stella octangulas and octahedra is known, and so are the cube, octahedron and stella octangula inside the rhombic dodecahedron. The window rhombus is one of the ten pieces of Lalvani's building system (a 1988 patent), but the Jewel is not: its walls lie in the stellas' planes, and the patent has no stella octangula or carving. We did not find a cube built from carved dodecahedra, or this block of eight, in a web search or in DICTO's reference library. As the paper says, this is not proof of novelty, and the record makes no claim of priority. The Hexa can be built in Polyhedraverse and seen in Kaleidohedra's Stella–Jewel world.

Data: Appendix A.18 (volumes, splits, clusters and the 16 checks).

The Hexa and its names are DICTO's. Design study and checking with Claude (Anthropic) in the open-source library krp-core. First released inside Kaleidohedra v2026.10.11-octet. From paper #18, Version 2, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23376320

AXE, CLEO and FUJI

The three gap pieces of the icosahedron

If you held them

You would hold three small pieces. **AXE** is shaped like an axe head: two equilateral triangles hinged together, with the sharp edge as the blade. **CLEO** is a slender pyramid, named after Cleopatra's Needle, the obelisk in London. **FUJI** is a hexagonal pyramid with the shape of Mount Fuji; its colour is that of the fuji flower, Japanese wisteria.

FUJI's base is a hexagon with all its angles equal, but with long and short sides taking turns. All six of its slanting edges have the same length.



Figure 19. AXE, CLEO and FUJI, at the one scale at which they fill the Lotus seed. From the paper's Figure 1.

The problem they solve

Twelve balls fit snugly round one, and their twelve centres are the corners of an icosahedron. Frank showed in 1952 that this cluster of thirteen atoms is more stable than the matching pieces of ordinary crystals. But it cannot be continued through space: icosahedra placed round each other always leave gaps. Two icosahedra meeting along an edge leave a wedge of empty space on each side of it.

DICTO's cluster, the **Lotus seed**, places twelve icosahedra round a small one, and fills every gap with just three kinds of piece. Thirty AXE close the outer grooves and thirty more the inner ones. Twenty FUJI fill the outer corners, where three icosahedra meet. Twenty CLEO fill the inner corners round the small centre. The result, in 113 pieces, is exactly one large regular icosahedron.

We first made it in the Polyhedraverse app on 5 October 2026, while working on a dodecahedron problem, in a simpler way: a dodecahedron, an icosahedron-with-a-cap-missing (a J11) on each of its 12 faces, then 30 AXE and 20 FUJI. That is 63 pieces, and the easiest way to build the Lotus seed by hand. The paper comes with printable nets for it.

What we found

AXE and CLEO turned out to be two of the Mosseri–Sadoc tiles. Sadoc and Mosseri made those tiles to describe icosahedral order in real materials. Here the same tiles arrive on their own, as the gap pieces of a cluster built from whole icosahedra.

FUJI is new. We prove that it cannot be cut up into Mosseri–Sadoc tiles of its own size or larger. One proof uses the Dehn invariant, a quantity that cutting and gluing cannot change. The other uses the angles of FUJI's hexagonal base. Whether smaller tiles could make it is left open.

The three pieces turn up all through DICTO's work. Add 20 CLEO to a small icosahedron and you get Kepler's great stellated dodecahedron. Add 30 AXE and you get the excavated dodecahedron. Add 12 caps and you have the dodecahedron: the Dodecahedral Chain, from the icosahedron to its partner through two star solids. The Chain of Karma (#22) repeats the Lotus seed at every scale, and AXE, CLEO and FUJI stay whole at every step.

What it adds

R. W. Gray described the cage of twelve icosahedra in 2002, and noted that they outline a larger one; he did not fill the gaps. AXE and CLEO are the Mosseri–Sadoc tiles. Koca and colleagues note that those tiles alone cannot grow the icosahedron past one golden step. With whole icosahedra and FUJI in their corners, the Lotus seed goes further, and the Chain of Karma carries on without end.

Data: Appendix A.19 (the pieces' edges and volumes, the angles, the two fillings, the FUJI proofs).

The Lotus seed, the 63-piece build, UNITY, the Chain of Karma, the Stella-Corona and the names AXE, CLEO, FUJI, Hasu, Triskelion, Triskelion Hex, Inner Eye, DESHI and King of Pentacles are DICTO's. AXE and CLEO are the Mosseri–Sadoc tiles t6 and t4; the DESHI arrangement is R. W. Gray's. Computation, checking and drafting with Claude (Anthropic). From paper #19, Version 3, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23351291

The FUJI Family

Pyramids on flat-ended triangles, and a new pair that fills space

If you held one

You would hold a pyramid on a six-sided base. The base is what DICTO calls a **flat-ended triangle**: a triangle with its three corners cut off flat, so that long and short sides take turns and every angle is the same. All six slanting edges of the pyramid are equal. Three of its sides are equilateral triangles; the other three are narrower.

Now imagine a slider for the short sides. Shrink them to nothing and the pyramid is a regular tetrahedron. Let them grow to match the long sides and the pyramid sinks flat into a regular hexagon. Between them, at a golden setting, is FUJI, the gap piece of the Lotus seed (#19). That slider is the FUJI family.

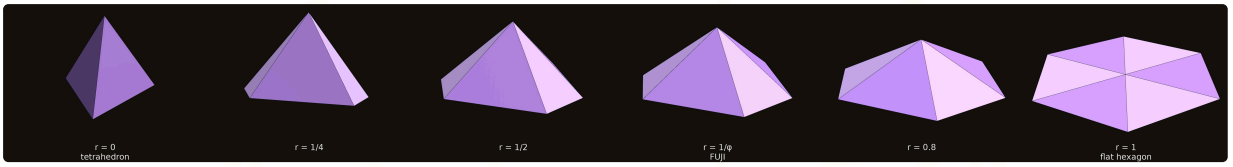


Figure 20. The FUJI family at one scale, all side edges equal: from the tetrahedron through FUJI to the flat hexagon. Darker faces: the three equilateral triangles; lighter: the three isosceles triangles. From the paper's Figure 2.

Where it started

The shape kept turning up. Designing the dictospheres, 3D-printed spheres with 14 PET-bottle ports, DICTO wanted sharp triangles round the ports and settled for blunted, flat-ended ones, to keep an even margin round every port. Colouring the Lotus seed, he hoped for regular hexagons and found the same flat-ended triangles. His DeltaZip and B1NR sculptures use rounded triangles, the curved form of the same idea. So we asked whether FUJI belongs to a family. It does.

What we found

Each antiprism picks a member. An antiprism is a drum of triangles between two equal polygons. For each one there is exactly one family member whose base carries on flat from the antiprism's triangles. The octahedron gives the tetrahedron. The square antiprism gives a “silver” member, tied to the silver ratio. The pentagonal antiprism, the middle of the icosahedron, gives FUJI, tied to the golden ratio.

The members live in many places. FUJI's base is the golden hexagon of a construction after George Odom. The truncated tetrahedron that Damasceno, Engel and Glotzer found to fill space has another member's base as its faces. In a “breathing” pyrochlore, a crystal of large and small tetrahedra found in some chromium oxides, every hexagon is a flat-ended triangle; at the golden setting, every hexagon is FUJI's base. And when a cube

and an octahedron are cut to the same depth, the solid's eight triangles belong to the family, with the cube's corners exactly at the pyramids' tips.

Wheels. Inspired by DICTO's DeltaZip wheels, we asked how many flat wheels can cross, each pair along a different spoke. The answer is at most three square wheels, four hexagonal ones or six ten-sided ones; five eight-sided wheels are impossible. Six wheels lead, by a roundabout route, to a sphere of 20 CLEOs and to the icosahedron cut into 20 pyramids from its centre, the **Cleo-Icos**.

A new pair that fills space. The Cleo-Icos cannot fill space on its own. But it is exactly the sharp corner of a golden rhombohedron, a slanted box whose six faces are golden rhombi. Cut that rhombohedron twice, and it falls into a Cleo-Icos at each sharp tip and a middle piece that DICTO named **Antonius**, four times as large. Golden rhombohedra fill space, so **Cleo-Icos and Antonius fill space, two to one**, face to face. The icosahedron's own pieces, with one partner, fill space. The same pieces build the rhombic hexecontahedron. We checked a patch of 2,187 pieces with no gaps or overlaps.

What it adds

Much here is known and credited: the hexagon shape, the antiprism angles, the golden hexagon after Odom, the space-filling truncated tetrahedron, breathing pyrochlores, and the Cleo-Icos itself, which is Mackay's unit tetrahedron and one of the pyramids in Hypsicles' “Book XIV” of Euclid, about 150 BC. Our searches did not find the family as one object, the antiprism rule, the wheel bound, or the split of the golden rhombohedron into Cleo-Icos and Antonius; the paper reports what was searched and does not claim these are unknown. The family gives one shape and one slider to things found separately: a gap piece, a space-filler found by computer, the hexagons of magnetic crystals, and the faces of a cut cube.

Data: Appendix A.20 (the base-angle formulas, the antiprism rule and its proof, the table of members, the wheel bound, Cleo-Icos and Antonius).

FUJI, Cleo-Icos, Antonius, the Lotus seed, the dictospheres, the DeltaZips and B1NR sculptures, the term “flat-ended triangle” and the names are DICTO's. Computation, checking and drafting with Claude (Anthropic). From paper #20, Version 5, 20 October 2026 (text of Version 4). Licence CC BY-SA 4.0.

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The DICTO Truncated-Icosahedral Hecatodihedron

A solid of 102 faces, found by building one stage of the Spheres of Dharma

If you held it

You would hold a nearly round solid with 102 flat faces of three kinds: 12 regular pentagons, 30 rhombi and 60 slim triangles. The pentagons sit a little lower than the rest, like shallow dimples.

It is built on the truncated icosahedron, with a low six-sided pyramid, a **Shield**, standing on each of its 20 hexagons. The Shields are raised to just the height where the triangles of two neighbouring Shields lie flat in one plane and merge into a rhombus. Its edges fold at only two angles.

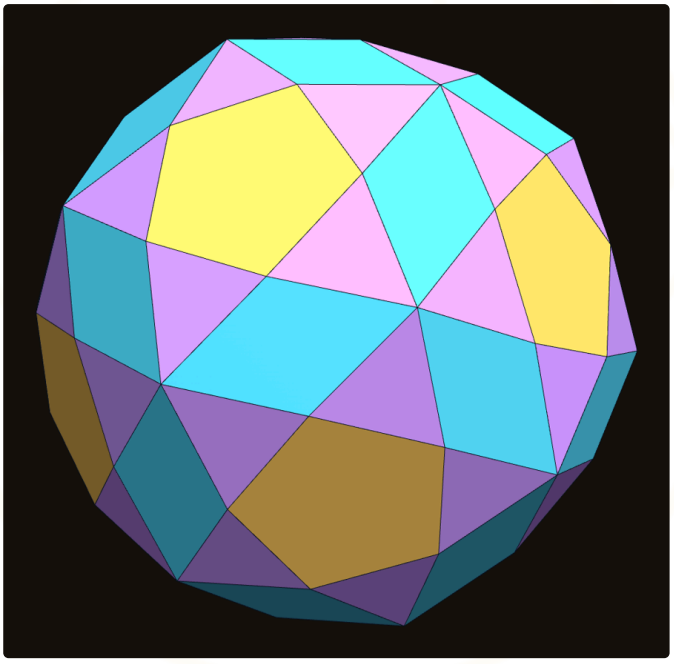


Figure 28. The Hecatodihedron: 12 regular pentagons (gold), 30 rhombi (cyan), 60 triangles (wisteria). From the paper's Figure 1.

Where it started

The Spheres of Dharma (#23) grow an icosahedron layer by layer, through the icosidodecahedron, the dodecahedron, the truncated icosahedron and the rhombic triacontahedron, back to an icosahedron a golden step larger. In *Capturing the In-betweens* (#27) each closed stage of that growth is taken as one solid. Stage 7, the truncated icosahedron with a Shield on every hexagon, turned out to be convex, with fewer faces than its pieces suggest: 102. It was found by building it.

What we found

At the special height, the 60 triangles across the edges where two hexagons meet pair up into 30 rhombi. The 60 triangles next to the pentagons stay triangles. The 20 tips of the Shields then sit exactly on the corners of a regular dodecahedron.

The rhombi and the triangles, 90 faces in all, touch one sphere. The 12 pentagons lie a little inside it, so the solid has no sphere that touches every face. An earlier paper of ours said all 102 faces touch one sphere; that was wrong, and it has been corrected.

What the solid is, exactly: the tips of the Shields are the corners of a known solid, the pentakis dodecahedron, the partner of the truncated icosahedron. So the Hecatodihedron is a known solid, the joined truncated icosahedron, with its 12 five-fold corners cut off flat at the pentagons: an unnamed solid derived from a known one. Its partner, made by turning it inside out, is the DICTO Ogdoecontahedron (#31). The two interlock into one compound of 182 faces, which DICTO named the DICTO Hecatogdoecontahedron (#34a).

What it adds

Its pattern of faces is close to Conway's hexakis truncated icosahedron, which has 132 faces; ours is that solid flattened, with 30 pairs of triangles merged. It is not an Archimedean, Catalan or Johnson solid. We searched Conway's notation, the Johnson solids and their near misses, MathWorld, the Polytope Wiki and McCooey's catalogue, and did not find it named. That is a report of where we looked, not a claim that the solid is unknown. The name is DICTO's: *hecatodi-* for its 102 faces, from the Greek for one hundred and two, as in the hecatonicosachoron; truncated-icosahedral for the solid it grows from. It can be opened in Polyhedraverse (DICTO Transitional Shells) and seen in the DICTO-TIC world.

Data: Appendix A.28 (faces, distances, angles, volume, the exact relation to the joined truncated icosahedron, the searches).

The solid is a stage of DICTO's Spheres of Dharma, found by building it, and the name is DICTO's. Computation and checking with Claude (Anthropic). From paper #28, Version 3, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23360857

The DICTO Ogdoecontahedron

A solid of eighty faces, found by turning a set of nested shells inside out

If you held it

You would hold a nearly round solid with 80 flat faces: 20 regular hexagons and 60 kites. Everywhere you look, kites crowd round the hexagons. At 12 points, five kites meet in a point that stands just a little proud of the rest.

Roll it in your hand and it looks the same from many sides. It has the full symmetry of the icosahedron, the twenty-sided regular solid. Its 102 corners almost all lie on one sphere: 90 of them sit exactly on it, and only the 12 five-kite points sit just outside. The faces meet along edges of only two lengths, and they fold at only two angles, which are almost the same. Each kite has two long sides, the same length as the hexagon's sides, and two shorter ones. DICTO named it for its eighty faces: **Ogdoecontahedron**, from the Greek words for “eighty” and “face”. For short it is the OG.

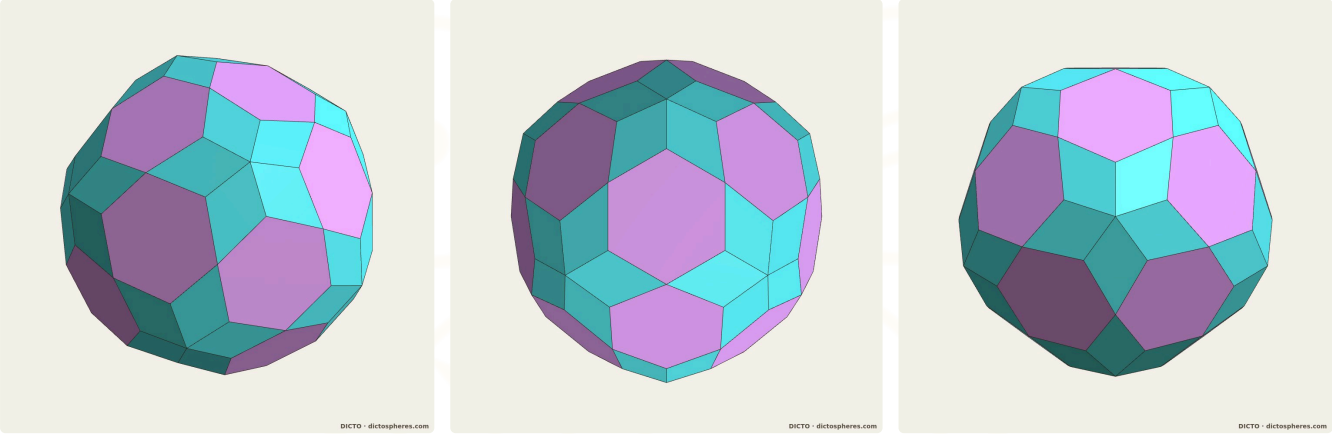


Figure 31. The DICTO Ogdoecontahedron, hexagons wisteria, kites cyan: (a) seen at a slant; (b) looking straight at a regular hexagon in the centre; (c) looking straight at one of the 12 outer points where five kites meet. From the paper's Figure 1.

How it was found

It began with an earlier work, the *Spheres of Dharma* (#23). There, six solids sit inside each other like Russian dolls. The gap between each doll and the next is filled exactly by a few repeated pieces, and the nesting repeats without end. Part-way through one layer, the caps on its hexagons close into a solid of their own, the DICTO Truncated-Icosahedral Hecatodihedron (#28).

On 13 October 2026 it was DICTO's idea to turn the whole nesting inside out. Simply turning each layer's pieces over does not work: no layer closes into a shell. What does work is an old trick of geometry. Take one sphere round the common centre and use it like a mirror: every flat face becomes a corner, and every corner becomes a flat face. Each solid turns into its partner, and the order of the nesting reverses, so that the inside goes outside.

We did this to all seven solids: the six of the nesting and the Hecatodihedron between them. Six came out as solids that were already known. One did not. The Hecatodihedron turned into this new solid, the Ogdoecontahedron.

What we found

The OG has a simple description once it is found. Take a known solid, the pentakis dodecahedron. Find its 20 corners where six faces meet, and slice each one off flat. Each slice leaves a regular hexagon, and what remains of the old faces are the kites. Adding a cap on one side of the mirror is cutting a corner on the other: the Hecatodihedron adds a low pyramid to each hexagon of its parent, and the OG cuts a corner instead.

In the reversed nesting the OG sits snugly between two classical solids. Its hexagons lie in the faces of the icosidodecahedron inside it, and its kites lie in the faces of the pentakis dodecahedron around it.

It was DICTO's idea, too, to join copies of the OG hexagon to hexagon. Join each copy to four others, through four hexagons that point like the corners of a tetrahedron, and the copies form a diamond network. The joints themselves sit on a pattern of corner-sharing tetrahedra, the pyrochlore lattice. A second network fits into the empty places of the first, touching it only on hexagons. Together the two make one tidy stack: every copy faces the same way and touches eight neighbours, through the eight hexagons that point to the corners of a cube. This stack fills 77.43 per cent of space. That is more than the tightest possible stack of equal balls, like oranges on a market stall, which fills 74.05 per cent. A computer search found a slightly denser regular stacking, but it is not proved to be the densest.

The OG joins the regular hexagon, at home in repeating patterns, to five-fold symmetry, the symmetry of quasicrystals. We tested whether it is a piece of the standard icosahedral quasicrystal. It is not. But it can be stacked in five such ways, and where the five stacks meet round a five-fold axis, the copies that must be taken out, with the copy in the middle, form exactly the Fibonacci chain near the centre. That is the subject of *A Fibonacci Spine* (#30).

What it adds

Much of this is old and is credited in the paper: turning solids inside out about a sphere, the pentakis dodecahedron and the cutting of its corners, the diamond and pyrochlore lattices, and the stacking of equal balls. Even the OG's pattern of faces, which face meets which, is known: it can be written in the notation of the Antiprism software. What we did not find, in DICTO's reference library, in McCooey's online catalogue of 553 solids or in web searches, is the solid itself, with these regular hexagons and these kites, or its exact hexagon-joined stacks. So the OG is an unnamed solid derived from a known one. Its name is DICTO's; the solid with these faces, and its stacks, were not found in the sources checked.

Data: Appendix A.31 (counts, exact edge lengths, angles, radii and volume; the stacks and their density; the quasicrystal tests; the sources searched).

The Hecatodihedron, the idea of turning the Spheres of Dharma inside out, the name Ogdoecontahedron and the idea of joining copies on their hexagons into pyrochlore networks are DICTO's. Computation, exact arithmetic and renders with Claude (Anthropic). From paper #31, Version 4, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23404688

The DICTO OCD Stella

The Ogdoecontahedron's companion, cut along its joints

If you held it

You would hold a faceted, gem-like solid with 48 corners and 50 faces: 6 hexagons and 24 four-sided faces, all set in golden planes, and 20 triangles. Eight of the triangles are special. They are the flat places where it meets its neighbours.

It is not as round as the icosahedral solids around it. Its cut keeps only 24 of their symmetries, the ones they share with a cube. (Geometers call this pyritohedral, after the crystals of pyrite.)

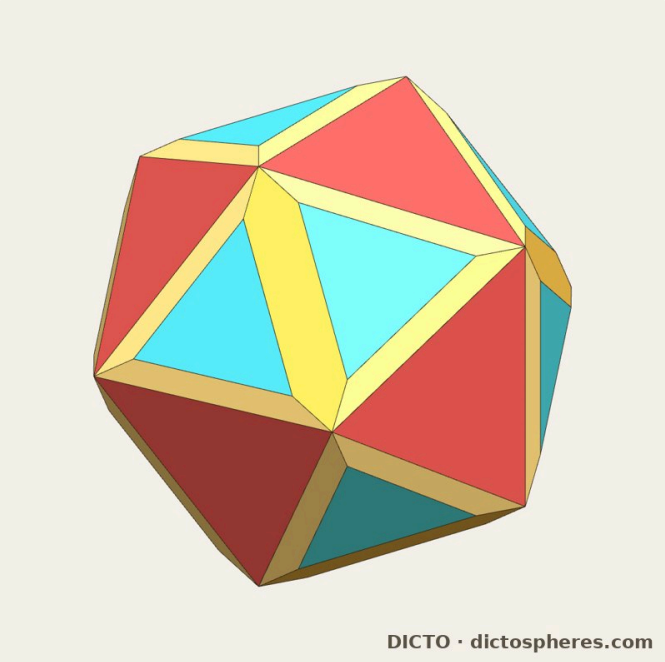


Figure 31a. The DICTO OCD Stella. From the paper's Figure 1 (left).

Where it started

The DICTO Ogdoecontahedron (#31) packs in a body-centred cubic arrangement, each copy joined to eight neighbours on its hexagons. Golden adapter pieces fit round each copy, and together they have a solid outline, the OG's **companion**. In DICTO's words, it was “a free gift from the way the adapters sat on the OCD”. Cut the companion along the eight joints, and you have the **DICTO OCD Stella**.

What we found

It packs. Copies of the OCD Stella meet face to face across all eight joints and fill 87.12 per cent of space.

It makes a seed. Set twelve OCD Stellas round a small one, and they fill a larger OCD Stella exactly, the same pattern as the Bodhi Seed (#24), even though the OCD Stella is

less symmetric.

It makes crystals. The OG's packing splits into two diamond networks. It was DICTO's idea to put a partner on the second network while the OCD Stellas stay on the first. With icosahedra as partners, every solid meets its eight neighbours face to face. It was DICTO's idea, too, to cut the remaining gap into pieces, and it falls into exactly four kinds of convex piece. With regular octahedra as partners, DICTO's next idea, each octahedron's six tips nest in notches in the OCD Stellas, and the crystal fills 96.33 per cent of space. With octahedra on only a quarter of the sites, as in rock salt, it fills 91.72 per cent.

The Atom 13. DICTO's idea again: twelve OCD Stellas round a central one, each pushed out until it touches the centre. They meet the centre flat and never touch each other. Twenty icosahedra fit into the gaps between them.

What it adds

The companion's pattern of faces is known (the chamfered icosahedron), and a different golden space-filler cut from the rhombic triacontahedron on the same kind of lattice is known (Kabai's 54-faced space-filling polyhedron). That one fills all of space; the OCD Stella is a different solid and fills 87.12 per cent. Crystals of two kinds of hard solids, such as tetrahedra with octahedra, are also known. We did not find the OCD Stella, its packing, its seed, its crystals, the gap pieces, the notched OCD Stella or the Atom 13 in the sources checked, which included 64 texts of DICTO's reference library searched in full. You can see the crystals in Kaleidohedra (3D+ → OCD Stella Crystals).

Data: Appendix A.31a (the solid's exact volume and faces, the seed, the four crystals and their fills, the gap pieces, the Atom 13).

The companion, the stella cut, the seed, the name, the partners on the second network (icosahedra and octahedra), the pyrochlore arrangement, cutting the gap into pieces and the Atom 13 are DICTO's; the exact computations and the text were done with Claude (Anthropic). From paper #31a, Version 3, 20 October 2026.
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DICTO DOI: 10.5281/zenodo.23430870

The DICTO Hecatogdoecontahedron

Two partner solids locked into one

If you held it

You would hold two solids passing through each other, like the two tetrahedra of a star-shaped stella octangula. One is the DICTO Hecatodihedron (#28), with 102 faces. The other is the DICTO Ogdoecontahedron (#31), with 80 faces. Corners of each poke out through the faces of the other.

Together they have 182 faces and 182 corners. DICTO named the pair by merging the two names: the **Hecatogdoecontahedron**, 102 and 80 together.

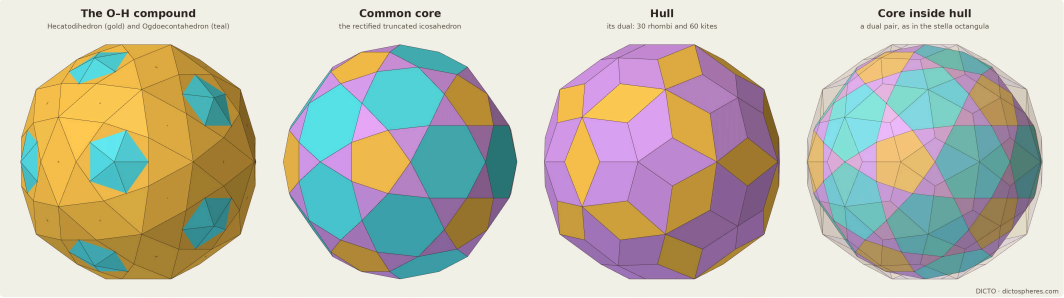


Figure 34a. The DICTO Hecatogdoecontahedron: the Hecatodihedron (gold) and the Ogdoecontahedron (teal) interlocked, with the truncated icosahedron and the common core and hull. Drawn from exact corners. From the paper's Figure 1.

What we found

The two solids are partners: turn one inside out about a sphere, and you get the other. Here the sphere is the one that touches the middle of every edge of the truncated icosahedron. At that size the two solids interlock exactly. The Ogdoecontahedron's 90 corners on that sphere sit exactly at the 90 edge midpoints of the truncated icosahedron. No corner of either solid lies inside the other.

Because the two are partners, the pair is its own partner: turned inside out, the compound swaps its two solids and stays the same.

The part the two solids share, their common core, is the rectified truncated icosahedron. The outline that wraps them both, the hull, is the joined truncated icosahedron, also called the rhombic enneacontahedron. These are exactly the core and hull of a classical pair, the truncated icosahedron with its partner the pentakis dodecahedron. So the new pair shares out the space of a classical compound in a new way. It is like the octahedron and cube, which are the core and hull of the stella octangula.

In *Progenitors* (#34), the core and the hull become stations of a golden cycle, so that cycle holds the cores and hulls of two partner compounds: the classical icosahedron with dodecahedron, and this one.

What it adds

Compounds of a solid with its partner are known. So are the classical truncated icosahedron–pentakis dodecahedron compound, the rectified truncated icosahedron and the rhombic enneacontahedron. We claim only the pairing and the name, not the core or the hull. The compound of these two solids was not found in the sources checked, and the name returns no search results. It can be seen in Kaleidohedra (3D+ → Vajrayana, and Multi-shell chains).

Data: Appendix A.34a (the counts of the two solids, the compound, its core and hull).

The pairing and the name are DICTO's; the exact computations and the text were done with Claude (Anthropic). The compound was first described in Version 2 of the Hecatodihedron paper (#28). From paper #34a, Version 2, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23430868

The DICTO Metaxyhexecontahedron

The solid in the last gap of the golden cycle

If you held it

You would hold a solid of 60 identical triangles, slightly pointed at 12 places where five triangles meet and blunter at 20 places where three meet. It has 32 corners and 90 edges. A ball could sit inside it touching all 60 faces at once.

It is made by wrapping one skin round two classical solids at once: a regular icosahedron, and a dodecahedron whose corners sit over the icosahedron's faces. How big the dodecahedron is decides the shape. DICTO's solid is one exact size in that range. Its name means “the in-between solid with sixty faces”, from the Greek *metaxý*, in between.

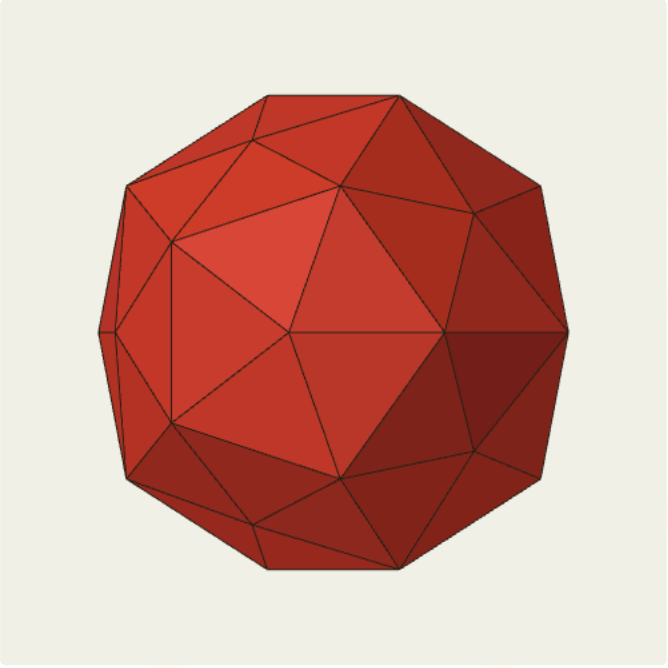


Figure 35a. The DICTO Metaxyhexecontahedron. Drawn from exact corners. From the paper's Figure 1 (left).

Where it started

The Spheres of Dharma (#23) nest icosahedral shells in a cycle that repeats at a golden step. *Progenitors* (#34) set the DICTO Ogdoecontahedron into it, and *Vajrayana* (#35) closed its loose steps, all but one. In the last step, from the rhombic triacontahedron to the next icosahedron, the triacontahedron reaches only part of the way. It was DICTO's idea that one more solid should fit there. No classical convex icosahedral solid touches both neighbours. This one does.

What we found

The Metaxyhexecontahedron fits the gap exactly. Its 12 five-fold corners are the triacontahedron's own corners, and its 20 three-fold corners sit exactly on the centres of

the next icosahedron's faces. It touches both sides.

Better still, each of the two layers it makes is filled by a single kind of piece: 30 identical pieces between the triacontahedron and the new solid, and 60 identical pieces between the new solid and the next icosahedron. Its partner, made by turning it inside out, closes the matching step in the inside-out version of the cycle.

It belongs to a family: all the skins round a dodecahedron and an icosahedron in this position, as the dodecahedron grows. That family holds the Catalan pentakis dodecahedron, the DICTO Palintriacontahedron (#35b) and the DICTO Metaxyhexecontahedron. One curiosity is noted without explanation: the size ratio that defines it is also the exact gain per round of the tightest cycle in *Progenitors*.

What it adds

The family is known: the Polytope Wiki gives its range, and MathWorld and McCooey give the Catalan member. Not found in the sources checked: this member as a named or distinguished solid, or its size ratio as distinguished. Its number matches no sequence in the OEIS, and the name returns no search results. It can be seen in Kaleidohedra (3D+ → Vajrayana).

Data: Appendix A.35a (the closed form, edges, radii, insphere, volume, the cycle step).

The idea that one more solid fits the last gap of the cycle, and the name, are DICTO's; the exact computations and the text were done with Claude (Anthropic). From paper #35a, Version 3, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23430864

The DICTO Palintriacontahedron

The rhombic triacontahedron turned inside out

If you held it

You would hold a solid of 60 triangles with 32 corners: 12 where five triangles meet and 20 where three meet. Each triangle is exactly half of a rhombus from Kepler's rhombic dodecahedron, the familiar twelve-sided solid of garnet crystals and honeycomb cells.

Its name, from the Greek *pálin*, “back”, says what it is: the rhombic triacontahedron turned back on itself. It points along the same 32 directions as the triacontahedron, but where that solid's corners are far out, this one's are near in, and the other way round.

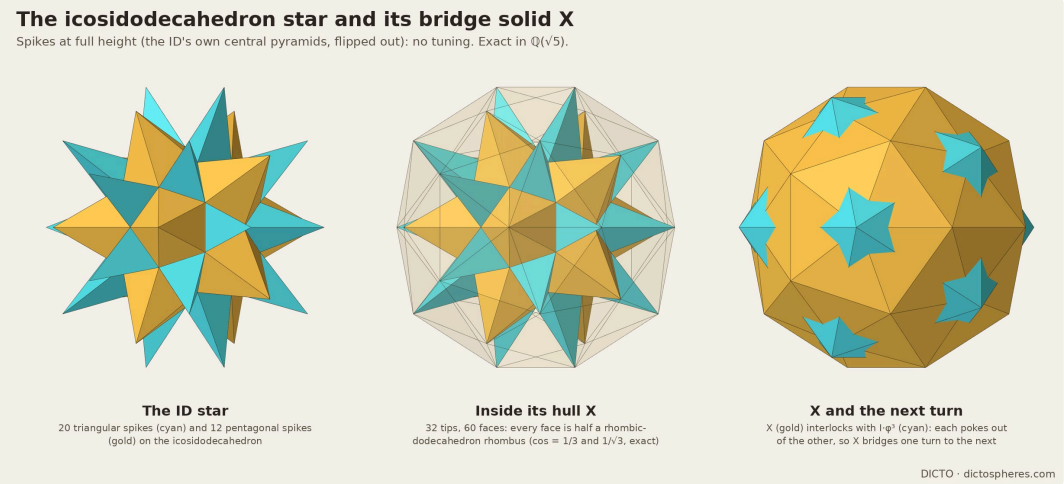


Figure 35b. The icosidodecahedron star (its central pyramids flipped out at full height), inside its hull, the DICTO Palintriacontahedron, and the Palintriacontahedron interlocked with the next icosahedron. Drawn from exact corners. From the paper's Figure 1.

Where it started

In the twelve-station golden cycle of *Vajrayana* (#35), the icosidodecahedron is the shell where the cycle's two progenitors meet. An icosidodecahedron can be cut into 32 pyramids from its centre, one over each face. DICTO flipped those pyramids outward, at their full height, rather than shortening them. In his words: “shortening feels like cheating; one more solid is needed”. The skin round the 32 tips is the Palintriacontahedron.

What we found

It is the triacontahedron turned inside out: the same 32 axes, with the ratio of its two kinds of corner distance reversed. It is also a member of the same family as the Metaxyhexecontahedron (#35a): the skins round a dodecahedron and an icosahedron. Its place in that family lies between the Catalan pentakis dodecahedron and the Metaxyhexecontahedron.

It reaches across the cycle. Measured by computer, it lies inside the next turn's dodecahedron, and almost all of it, 99.46 per cent, lies inside the next icosahedron's Cleo star. So it bridges one turn of the golden cycle to the next. A small coincidence of numbers: the first stellation of the icosahedron puts its tips at the same ratio that defines this solid.

What it adds

The family and its range are known (the Polytope Wiki), and so are the rhombic triacontahedron and the icosidodecahedron's central pyramids. Not found in the sources checked: this member as a named or distinguished solid, or its size ratio as distinguished. The name returns no search results.

Data: Appendix A.35b (the closed form, the face angles, edges, radii, insphere and volume).

Flipping the icosidodecahedron's own central pyramids out at full height, and the name, are DICTO's; the exact computations and the text were done with Claude (Anthropic). From paper #35b, Version 2, 20 October 2026.
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DICTO DOI: 10.5281/zenodo.23430866

The DICTO Jewel

Euclid's dodecahedron carved by its neighbours' stars

If you held it

You would hold a cut gem with 60 faces: twelve diamond-shaped windows and 48 small triangles. Each window is a thick Penrose rhombus, the fatter of the two tiles of Penrose's famous never-repeating pattern, with corners of 108° and 72°. On each of its six sides there is a small dimple.

Underneath it is a cube. The windows lie across the cube's twelve edges, and the triangles are the walls where it was carved.

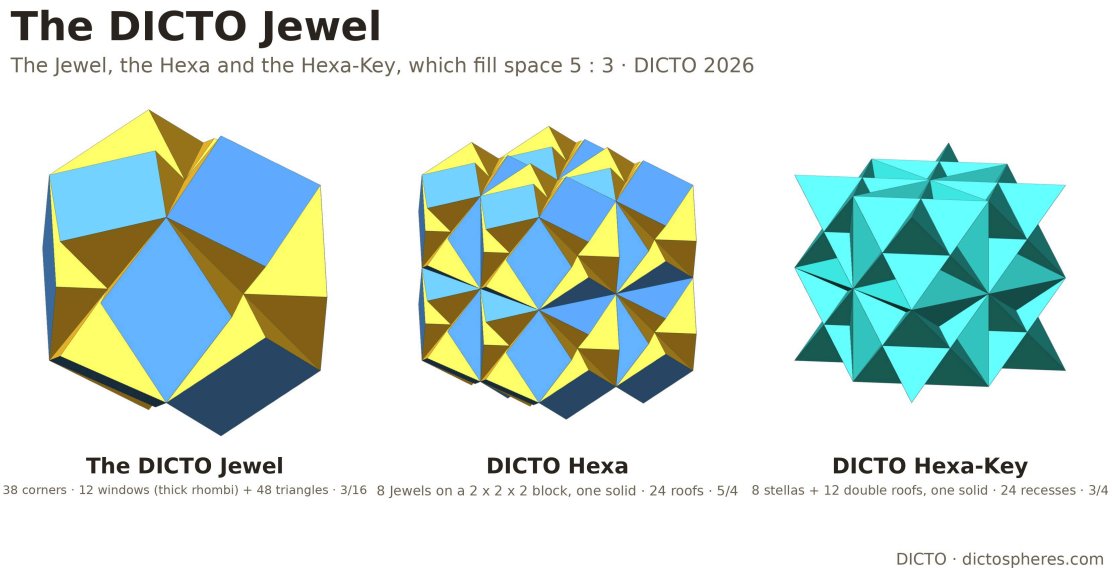


Figure 41. The DICTO Jewel. From the paper's cover figure.

How it is made

Euclid built the regular dodecahedron by setting a roof on each face of a cube. Now place cubes side by side to fill space, like a checkerboard in three dimensions. Put Euclid's dodecahedron on the white cubes. On the black cubes put the star made of two interlocking tetrahedra, Kepler's stella octangula. Each roof of a dodecahedron reaches into a black cube, and the star there carves part of it away. What is left is the **DICTO Jewel**: the cube and its six carved roofs.

What we found

Jewels and stars fill space. Jewels on the white cubes and stella octangulas on the black cubes fill space exactly, three parts to one by volume, in the pattern of table salt. This was first checked by sampling points; here it is proved exactly. Each black cube is precisely its star plus the six carved roofs pushed into it by its neighbours.

Clusters named as solids. DICTO's clusters of Jewels become single solids with his names. The **Jewel-Tetra** is four Jewels that all touch, window to window. The **Jewel-Octa** is six Jewels round a black cube, with that cube's star sealed in the hole between them, filling it exactly. The **Jewel-Rhombi** is eight Jewels on the corners of a slanted box, again with a star sealed inside.

How they fill space. Jewels make up only three-quarters of space, so no group of Jewels alone can fill it: stars must come too. Each of the three clusters fills space with copies of itself and loose stars, and we found every regular way of doing so. Together, Jewel-Tetras and Jewel-Octas sit exactly where the tetrahedra and octahedra of the octet truss sit, though as overlapping solids, not as a tiling.

The Hexa, proved. The DICTO Hexa and Hexa-Key of paper #18, and their five-to-three filling, are proved exactly here. The same pattern repeats one level up: a cube with roofs added is a Jewel and with roofs taken away a star; a block of eight with roofs added is a Hexa and with roofs taken away a Hexa-Key.

Twins. As a single solid, the Jewel twins like the DICTO Ogdoecontahedron. Its packing does not twin across a slanting mirror, and the paper gives the simple reason: that mirror does not keep the grid of cubes.

What it adds

The parts are classical: the roofed cube is Euclid's, the stella octangula Kepler's, the thick rhombus Penrose's, the arrangement that of table salt. The Jewel itself, its lattice and the clusters were first published by DICTO in the Kaleidohedra app and in *Kagome Hulls* (#17); the Hexa in #18. New here are the exact proofs, the three cluster solids and their face counts, every lattice tiling of the three with loose stars, the octet pattern with its limit, and the exact reason the packing does not twin. Lalvani's 1988 patent builds structures from pieces whose edges all lie on certain icosahedral axes; only the Jewel's window edges do, so it falls outside that condition. We did not find the Jewel, its clusters, the Hexa or the Hexa-Key outside DICTO's own records, in 80 library sources searched in full and in web searches.

Data: Appendix A.41 (the volumes, corners and faces, the tiling proof, the cluster counts and lattices, the Hexa proof, the twin).

The Jewel, its stella-octangula counterpart, its name, its clusters (and the Jewel-Tetra, Jewel-Octa and Jewel-Rhombi as single solids, and their names), the Hexa, the Hexa-Key and their names are DICTO's. Computation, exact proofs and drafting with Claude (Anthropic). From paper #41, Version 1, 20 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23496353

The Reuleaux Platonic Solids

Rounded bodies made where balls on the corners of a Platonic solid overlap

If you held one

You would hold a smooth, rounded body with curved faces meeting along curved edges, like a pebble that still remembers the solid it came from. Each face is a piece of the surface of one ball.

Oddly, each body looks like the partner of the solid it was made from. Balls on the corners of a cube make a body that looks like an octahedron, and balls on an octahedron make one that looks like a cube. So DICTO named them by their look: the **Reuleaux octahedron** (balls on the cube), the **Reuleaux cube** (on the octahedron), the **Reuleaux icosahedron** (on the dodecahedron) and the **Reuleaux dodecahedron** (on the icosahedron).

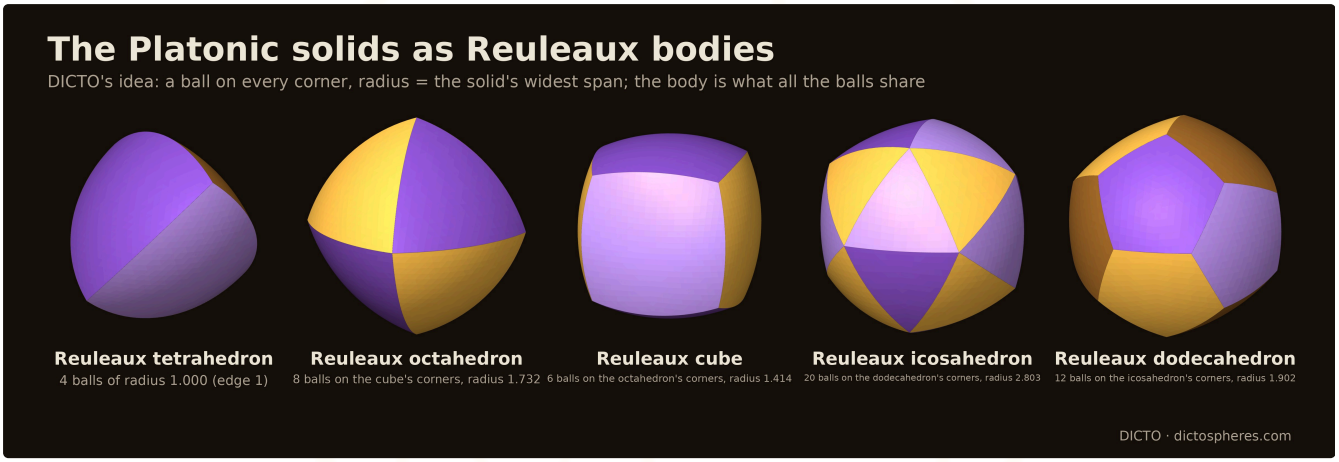


Figure 45. The five bodies at solid edge 1, each with its faces coloured with the fewest colours that keep neighbours apart (gold, purple, lilac, bronze). Under each: the number of balls and their radius, which is the solid's diameter. From the paper's Figure 1.

How they are made

It was DICTO's idea. Put a ball on every corner of a Platonic solid, each ball just large enough to reach the farthest corner. The space that all the balls share is a rounded body. On the tetrahedron this is the famous Reuleaux tetrahedron. Building the same body on all five Platonic solids, as one family, is DICTO's idea, and the other four did not appear in the sources we checked.

What we found

The main theorem says how thick these bodies are.

Never thinner than the solid is wide. A rounded body of the same thickness in every direction fits inside each one, and nothing can be thinner than a body it holds. Measured between two opposite corners, the thickness is exactly the solid's width.

Through a face, exactly. Measured straight through the middle of a face, the thickness is exactly a number we can write down. By symmetry the outermost point lies on that line, and it can reach only as far as the farthest corner's ball allows.

Partners agree. A solid and its partner, the cube and the octahedron, or the dodecahedron and the icosahedron, give bodies exactly as thick as each other through their faces.

The tetrahedron follows its own rule, and the classical Reuleaux tetrahedron comes out right: our computed volume agrees with the known formula. Despite their names, the four new bodies are not Reuleaux polyhedra in the strict mathematical sense: each corner lies on only one other ball's surface. They are “ball polyhedra”.

We believe the line through a face is also the thickest direction of all. The computer agrees to twelve digits, over 3,000 directions, but it is not yet proved. That is left as a conjecture.

Colours. The faces of each body meet just as the faces of the partner solid do. So the fewest colours that keep neighbouring faces apart are 4, 2, 3, 3 and 4, the numbers DICTO read from the look.

What it adds

The Reuleaux tetrahedron and its volume are known, as are ball polyhedra in general, Reuleaux polyhedra and Meissner's solids. The three library papers on these bodies use only the regular tetrahedron. In the 81 library sources searched in full we did not find the bodies on the other four Platonic solids or their widths. One paper on the subject, Sallee's, is in the library as a link only and was not read.

Data: Appendix A.45 (the theorem, the exact widths, the proof sketch, the table of the five bodies, the checks).

The construction (a ball on every corner of a Platonic solid, radius the solid's diameter) and the names by look are DICTO's idea; computation, proofs and drafting with Claude (Anthropic). From paper #45, Version 2, 22 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23529499

The Polyhedral Stretching Theorem

Slide a solid one edge-length, and every edge stays the same



If you held one

You would hold a long, faceted solid: a regular dodecahedron that has been stretched. Its twelve pentagons are all still there, at the two ends, and between them runs a belt of ten diamond-shaped faces. Every edge, on the pentagons and on the belt alike, has the same length.

Looked at end on, its outline is a ten-sided polygon. Looked at from the side, it shows the long hexagon where it all began.

Figure 49. The dodecahedron swept along the face diagonal of Euclid's cube to the next dodecahedron: twelve pentagons (violet) and a belt of ten long faces (lilac). Left: along the long edge, the long hexagon DICTO saw. Middle: end on, a decagon. Right: oblique. From the paper's Figure 1.

Where it started

In paper #47, dodecahedra sit on every other cube of a grid. Looking into that block in the Kaleidohedra app, DICTO saw a void, a long six-sided space, with two dodecahedra overlapping it on either side. It was his idea to make a dodecahedron variant to fit it. Taken together, the void and the two overlaps make one solid: the space a dodecahedron passes through as it slides from one of the two places to the other. DICTO had found a first member of this family earlier, also from overlapping dodecahedra: the stretched dodecahedron, published as finding F11.

What we found

The theorem. Take any solid whose edges are all the same length, such as a dodecahedron. Slide it one edge-length in a straight line, in any direction except along one of its own edges. Every edge of the shape it sweeps comes out one edge-length long. The reason is short: every edge of the new shape is either one of the solid's own edges, moved, or a copy of the slide itself, and both are one edge-length long.

What happens to the faces is easy to picture. The faces at the two ends stay just as they were. A face that lies along the slide gets longer and gains two sides. And round the middle grows a belt of diamonds, one for each edge on the solid's outline.

Special directions. Sliding along an axis of the dodecahedron's own symmetry gives the cleanest shapes. Slide through two opposite pentagons, and the belt has ten diamonds, all alike. The face-diagonal slide keeps all twelve pentagons.

Equal-edged and telescoped. DICTO names the whole family the **DICTO stretched solids**: equal-edged when the stretch is one edge, telescoped at any other length. The faces keep their kinds; only the belt grows.

Three and four caps. Instead of sliding along a line, put a dodecahedron at each corner of a triangle or a tetrahedron and wrap them together. Again every edge is equal, and equilateral triangles appear among the faces. DICTO also wondered whether turning one end might make the belt into triangles. It cannot: the rim zigzags, and the best band of triangles is still a third of an edge off. Three and four caps give the triangles instead.

Every equal-edged solid. The theorem works for all the Platonic solids. Stretch the tetrahedron and octahedron and famous shapes appear: the cuboctahedron, the truncated tetrahedron and three Johnson solids, a good sign that the stretch is a natural operation. The stretched icosahedra form a second, new-looking family.

Make your own. Printable nets come with the paper, and a telescoping model in three parts that slides from the short shape to the long one.

What it adds

The method is old: sliding a solid along a line is a standard construction (the Minkowski sum), and how faces behave under it is known in general. Hart and Picciotto built triangle-and-rhombus solids the same way, without pentagons, and for zonohedra the stretch of a zone is classical (Grünbaum, after Fedorov). We did not find the dodecahedron swept along a face diagonal, a body diagonal or a five-fold axis, nor the three- and four-cap forms, named or described. Since the construction is simple, they may exist somewhere unnamed; we present them as DICTO's family on that understanding. The stretch of the thirteen Archimedean solids is left for later.

Data: Appendix A.49 (the theorem in full, the proof, the tables of directions and of the icosahedron family, the checks).

The stretched dodecahedron (F11), the long hexagon seen in the dodecahedron layer, the three- and four-cap forms and the telescoping model are DICTO's ideas; computation, proofs and drafting with Claude (Anthropic). From paper #49, Version 2, 22 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23566671

The Delian Bond

Samson, Delilah and the octet between Kepler's rhombic dodecahedron and the cube

If you held them

You would hold two solids of exactly the same volume. Delilah, in blue, is a regular octahedron: eight equal triangles. Samson, in gold, is a cube with four of its corners cut off, every other corner, so deeply that he looks like a tetrahedron with its tips clipped.

Put them together and they fit: Samson's four triangles are exactly the size of Delilah's faces. Copies of the two fill all of space with no gaps. In each cube of the pattern there are eight Samsons and four Delilahs, and the Samsons form a star round a Delilah hidden at the centre. DICTO named the pair Samson and Delilah, from “Delian”.

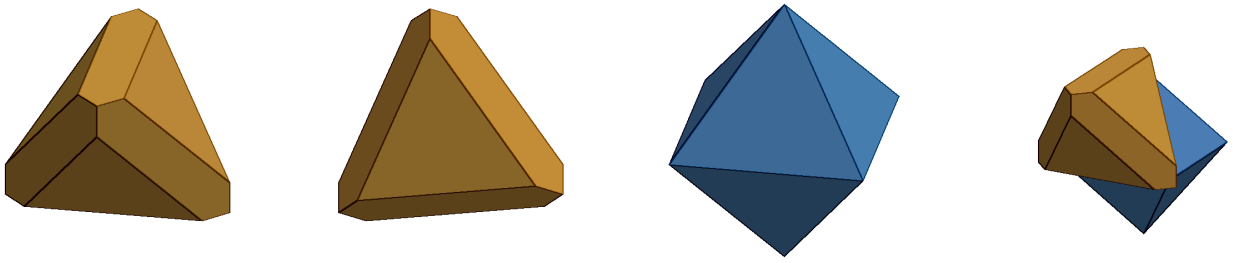


Figure 51. The Delian bond. From left: Samson (gold) from two sides, a cube with four alternate corners cut so deep that it looks like a clipped tetrahedron; Delilah (blue), a regular octahedron; the two face to face. Equal volume. Each panel at its own scale. From the paper's Figure 1.

Where it started

In paper #48, a single cut at the golden point turned a lattice of icosahedra and its holes into two convex solids that fill space together. On 23 October 2026 it was DICTO's idea to do the same for the three regular solids that build the octet: the tetrahedron, the octahedron and the cube. And, when a whole family of shapes appeared, his idea was to find its special positions first and name them afterwards.

What we found

Regular tetrahedra and octahedra fill space together, two tetrahedra for each octahedron. This classical pattern is the octet. Each of these solids can be seen as a bundle of cones, one from its centre over each face. Where a tetrahedron and an octahedron share a face, their two cones meet and make a double cone, and the double cones fill space too.

Now cut every double cone with one flat cut, parallel to the shared face, and give each side to its own centre. Slide the cut along, and the tetrahedra grow while the octahedra

shrink, or the other way round. It works like a slider. Wherever the cut is, space is still filled with two solids, and both stay convex, with no dents.

- At one end of the slider are Kepler's rhombic dodecahedra alone.
- Before the shared face, the pieces are a small regular tetrahedron and a rhombic dodecahedron with its eight corners where three faces meet cut off.
- A third of the way along is the octet itself.
- Past the shared face, the pieces are a regular octahedron and a cube with four alternate corners cut off.
- At the other end are cubes alone.

A few positions on the slider stand out. At two of them, one on each side of the octet, every edge in the whole filling has the same length. At another, the tetrahedron pieces and the octahedron pieces each fill exactly half of space. At no position is a hexagon regular.

And at exactly one position the two pieces have the same volume. That is the Delian bond. There Samson holds exactly twice the volume of the octet's tetrahedron, and Delilah exactly half the volume of the octet's octahedron: she is that octahedron shrunk. To double a volume you must stretch every length by the cube root of two. That is the Delian problem of the ancient Greeks, which cannot be solved with ruler and compass. Here it happens in both directions at once: one piece doubles while the other halves. The position also meets an earlier idea of DICTO's, doubling a solid's volume by the cube root of two.

The DICTONIC SOLIDS and BONDS

Paper #48 and this paper together give six solids that fill space in three pairs. DICTO calls the six the DICTONIC SOLIDS and the three pairs the DICTONIC BONDS: the golden bond (Icosaxes and Unborn Axes), the keyed bond (Lotus Jewel Axes and Dodecaxes) and the Delian bond (Samson and Delilah). Not all six are new as shapes. Delilah is a regular octahedron, and Samson is a cube with four corners cut, at a proportion we did not find in the sources. What is new in each bond is the bond itself: the two solids together, the cut that makes them, and the way they fill space as a pair. The same kind of cut tried on the cube meets only known solids, so the count stays at three bonds.

What it adds

The octet is old, and so are the links between the cube, the octet and the rhombic dodecahedron. Knoll and Morgan relate the three, but as separate steps. A general method, the weighted Voronoi or power diagram, gives families of such fillings, and this slider is one of them. Closest in spirit are the families of tetrahedra and octahedra found by Conway, Jiao and Torquato, and by Gabbrielli, Jiao and Torquato; but there both kinds of piece stay regular, while here one piece stays regular and its partner becomes a cut rhombic dodecahedron or a cut cube. What we did not find, in the sources we checked,

is the slider itself, from rhombic dodecahedra through the octet to cubes, its two equal-edged members, its half-and-half member, and the Delian bond. Still open: a single solid for each bond whose copies alone fill space, and printed sets of Samson and Delilah.

Data: Appendix A.51 (the slider and its exact volumes; the special positions; the Delian bond; the three DICTONIC BONDS; the cube and polar remarks; the checks and sources).

The ideas are DICTO's: to do for the tetrahedron, octahedron and cube what paper #48 did for the icosahedron, to find the special positions first and then name them, and the names Samson, Delilah, the DICTONIC SOLIDS and the DICTONIC BONDS. Computation, exact proofs and drafting with Claude (Anthropic). From paper #51, Version 1, 24 October 2026. Licence CC BY-SA 4.0.

DICTO DOI: 10.5281/zenodo.23606828

APPENDIX

The data, work by work

One section per work, with the exact values, volumes, proof sketches and checks as published in its paper. Exact values are given in the form the paper gives them; decimals (≈) are as printed there. Licence CC BY-SA 4.0.

A.18 The DICTO Hexa (DICTO DOI: 10.5281/zenodo.23376320)

The DICTO Jewel (scale: cube edge 2 per cell)

The regular dodecahedron of the Euclid–Kepler–Pacioli cell (Euclid's roofs on a cube of edge 2) carved by the stella octangulas of its six face-neighbour cells. It keeps the whole cube and 2/3 of each of its six roofs: volume exactly 12. Its surface keeps 12 rhombi of Penrose's thick 72°/108° shape, one on each cube edge (the windows), walled by 48 triangles in the stellas' planes. Jewels on the even cells and stella octangulas (volume 4) on the odd cells fill space: the Stella–Jewel Lattice, 3 : 1 by volume.

The Hexa and the Hexa-Key

- Hexa: eight DICTO Hexa Jewels on a 2 × 2 × 2 block, overlapping inside by 16 in total (8 × 12 = 96); one solid, a cube of edge 4 with 24 outward roofs, volume 64 + 24 × 2/3 = 80 (6⅔ Jewels). Inside, two tetrahedral Jewel clusters interpenetrate.
- No shear makes eight exact Jewels meet face to face as a cube: the lattice shear is symmetric, and no symmetric map takes the cube's edges to the Jewels' contact directions (even two of the three fail).
- Hexas mate only window to window, two whole windows per neighbour. At the FCC pattern of spacing 4: 12 neighbours, 24 whole windows shared.
- Hexa-Key: 8 stella octangulas and 12 double roofs, volume 48 (4 Jewels). Hexas and Hexa-Keys fill space as a checkerboard, 5 : 3 by volume (Jewels and stellas: 3 : 1).
- Split A: 8 cubes + 24 roofs (roof 2/3). Split B: 4 whole Jewels + 4 trimmed Jewels (3 roofs removed, volume exactly 8). Hexa-Key: 8 stellas + 24 roofs.

Clusters, networks, turns

- Rhombohedral cluster of eight Hexas: window to window, 36 whole windows shared, a Hexa-Key sealed inside, volume 688 = 57⅓ Jewels.
- Diamond cluster: Hexas sharing 7 corner Jewels, volume 556.
- Every Jewel in exactly two Hexas is impossible; best is the diamond network, each Hexa sharing 4 of its 8 corner Hexa Jewels with 4 neighbours, stella octangulas filling every cell between.
- No join in these packings can be turned by its face's symmetry and still fit; turns exist only for a piece hanging on one neighbour (on a window: 180° and one flip).

Checks

data/checks-hexa.txt: 16 checks from krp-core v0.16.0, rerun 15 October 2026, all passing: closed and outward surfaces; exact volumes (Hexa 80, Hexa-Key 48, clusters 688 and 556, trimmed Jewel 8, roof 2/3); 24 whole Jewel windows and stella-matching walls on the Hexa; every parts view adding up. Network, packing and turning results from the design study (overlap by sampling, exact contact areas, the mating scan and the turning table).

Sources

Known: the space filling by stella octangulas and octahedra; the cube, octahedron and stella octangula inscribed in the rhombic dodecahedron; the FCC primitive cell as a cube squashed along a diagonal. Lalvani, US 4,723,382 (1988): the window rhombus is one of its ten pieces; the Jewel's 48

walls lie 22.24° off every icosahedral two-fold axis. Pearce, US 3,974,600 (1976): struts with panels, not solid pieces. Not found (web search 9 October 2026; reference library 15 October 2026): a cube built from carved dodecahedra, or this block of eight. Not proof of novelty; no claim of priority. Also: Euclid, Elements XIII.17; Coxeter, Regular Polytopes.

A.19 AXE, CLEO and FUJI (DICTO DOI: 10.5281/zenodo.23351291)

Units: icosahedra of edge 1; $\varphi = \tau = (1 + \sqrt{5})/2$. Every volume, angle and count recomputed from the study data (CHECKS.json).

The pieces

Piece	Shape	Edges	Volume	Credit
AXE	tetrahedron	five of length 1, one of $1/\varphi$	1/12	Mosseri–Sadoc t6 at scale $1/\tau$
CLEO	triangular pyramid	base equilateral $1/\varphi$, side edges 1	$(\sqrt{5} - 1)/24$	Mosseri–Sadoc t4 at scale $1/\tau$
FUJI	hexagonal pyramid	base edges 1 and $1/\varphi$ alternating, six side edges 1	$(1 + 3\sqrt{5})/24$	new

- AXE: two equilateral triangles hinged at 41.81° (cos = √5/3). Two icosahedra sharing an edge leave $360° - 2 \times 138.19° = 83.62°$, two gaps of 41.81°.
- CLEO: faces meet at 63.43° along its long edges.
- FUJI: base angles all 120°; corners on a circle of radius √(2/3); apex height 1/√3.
- All dihedral angles from $\alpha = \arccos(\sqrt{5}/3) = 41.81°$ and $\delta = 37.38°$: $41.81° = \alpha$; $37.38° = \delta$; $63.43° = \pi - \alpha - 2\delta$; $79.19° = \alpha + \delta$; $116.57° = \alpha + 2\delta$ (the dodecahedron's); $142.62° = \pi - \delta$ (FUJI's side edges). Relations hold to 120 digits (integer-relation search).

The Lotus seed

- $I(\varphi^2) = 12\ I(1) + I(1/\varphi) + 60$ AXE + 20 FUJI + 20 CLEO: 113 pieces, no overlaps, volume $(235 + 105\sqrt{5})/12 = \varphi^6\ V(I)$. Each of its 20 large faces is three icosahedron faces and one FUJI base.
- $I(\varphi^2)$ = dodecahedron + 12 J11 + 30 AXE + 20 FUJI: 63 pieces.
- The 12 icosahedra share 30 whole edges; their inner rings are the 12 faces of a dodecahedron of edge 1 (the DESHI cage; Gray 2002).
- Triskelion (CLEO + 3 AXE) has the volume of the cap J2, $(5 + \sqrt{5})/24$. Inner Eye = $I(1/\varphi)$ + 30 AXE + 20 CLEO, volume $5\varphi/2$.
- $D(1) = I(1/\varphi) + 12$ J2 + 30 AXE + 20 CLEO (the Dodecahedral Chain through the great stellated and excavated dodecahedra).
- Chain of Karma steps 1–3: 113, 1,569 and 20,497 pieces. Stella-Corona volume $(120 + 47\sqrt{5})/3$.

FUJI is not made of Mosseri–Sadoc tiles

- FUJI has exactly the volume of 3 AXE + CLEO + t1 at scale $1/\tau$, but a different Dehn invariant. In (α, δ): FUJI 1-part $(3, -15/2)$, √5-part $(0, 3/2)$; 3 AXE + CLEO + t1′ 1-part $(27/2, 15/2)$, √5-part $(-1/2, 1/2)$. The √5-parts differ by $(\alpha + 2\delta)/2 = (\pi - \arctan 2)/2$; $\cos^2(\arctan 2) = 1/5$, so arctan 2 is not a rational multiple of π.
- Other mixes (all six tiles, four scales) with FUJI's volume and Dehn invariant are ruled out by the base: each 120° corner must be 60° + 60°, so the base must hold side-1 triangles on the long edges and side-$1/\tau$ triangles on the short edges; their total area 1.795 exceeds the hexagon's 1.669.
- Mixes of smaller tiles: open. CLEO fits inside FUJI and can grow to 1.023 times its size there.

Nets and sources

Nets print on A4 at 45 mm per unit (Lotus seed edges about 118 mm), each checked (faces congruent, every edge a fold or numbered glue pair, no overlap). References: Frank 1952; Mosseri & Sadoc 1982; Sadoc & Mosseri 1999; Kramer 1982; Koca et al. 2020; Gray 2002; Coxeter et al. 1938; Wenninger 1971; Dehn 1901; Kepler 1619. Scripts: studies/2026-10-10-icosa-13 and studies/2026-10-14-axe-cleo-fuji.

A.20 The FUJI Family (DICTO DOI: 10.5281/zenodo.23357896)

The family FUJI(r)

- Base: a flat-ended triangle with sides 1 and r (an equilateral triangle of side 1 + 2r with corners of side r cut off; the isogonal hexagon or truncated triangle t{3}); corners on a circle of radius² (1 + r + r²)/3.
- Side edges 1 when h² = (2 − r − r²)/3. Volume (√3/12)(1 + 4r + r²)h. Side faces: three equilateral, three isosceles 1, 1, r.
- Base angles: cos L = (1 + 2r)/3 (long sides); cos² S = (2 + r)/(3(2 − r)) (short sides). Proof: long-side distance s/(2√3) with s = 1 + 2r and slant √3/2; cut-line distance (2 + r)/(2√3) and slant √(1 − r²/4).

r	member	height	L	S
0	regular tetrahedron	√(2/3)	70.53°	– (limit 54.74°, no face)
1/4		3/4	60°	49.11°
1/2		0.6455	48.19°	41.81°
1/φ	**FUJI**	1/√3	41.81°	37.38°
1	flat	0	0	0

cos² L = 5/9 exactly at r = 1/φ, and cos² S = 5/9 exactly at r = 1/2: the 41.81° gap angle of the Lotus seed.

Odom, and the antiprism rule

- Extending each side of an equilateral triangle by e at both ends gives a flat-ended triangle with sides e and 1 + e, r = e/(1 + e); e = φ gives FUJI's base (Kabai 2008, after Odom); e = 1 gives r = 1/2. A change of parameter, not a new result.
- Face-plane condition L + D_X = 180°: r = (−3 cos D_X − 1)/2. Icosahedron (138.19°): 1/φ; cuboctahedron, triangle–square (125.26°): (√3 − 1)/2; square antiprism (127.55°): √2 − 1; rhombicuboctahedron (144.74°): (√6 − 1)/2; octahedron (109.47°): 0.
- Proposition 1: for the n-gonal antiprism, r = 2 cos(π/n) − 1, so 1 + r is the short diagonal of the regular n-gon. Proof: corners T_k, B_k with θ = π/n, R = 1/(2 sin θ); $(T_1 - M)\cdot(B_{-1} - M) = -(4\cos\theta - 1)/4$, so cos D_n = (1 − 4 cos(π/n))/3 (the standard antiprism dihedral, in the OEIS). Numerical check n = 3 to 12 to 10^{-12}.
- n = 3: tetrahedron; n = 4: √2 − 1 (inverse silver ratio); n = 5: 1/φ, FUJI; n = 6: √3 − 1. Flat-ended k-gons give families only for k = 3, 4 and 5 (flat at r = 2 sin(π/k − π/6)); the square family's rhombicuboctahedron member is r = 1 − 1/√2. Link to octagonal quasicrystals left open.

Where the members live

- Damasceno, Engel and Glotzer (2012): the truncated tetrahedron at t = 1/2 fills space; its hexagons have sides 2 : 1, the r = 1/2 base.
- Breathing pyrochlore (edges 1 and r): every kagome hexagon has sides 1 and r; with two kinds of truncated tetrahedra it fills space for every r (Wythoff construction in $\tilde{A}_3$); checked numerically at r = 0.2, 0.382, 0.5, 0.618, 0.8, 1. r = 1/φ: the golden breathing pyrochlore.

- Equal-depth cube–octahedron solid ($|x|, |y|, |z| \le 1$, $|x| + |y| + |z| \le \sqrt{3}$): 6 squares, 8 flat-ended triangles $r = (\sqrt{3} - 1)/2 = 0.366$, all 14 faces on one insphere; its cube corners are the apexes of FUJI($(\sqrt{3} - 1)/2$) scaled by $\sqrt{2}(\sqrt{3} - 1) \approx 1.035$. Proof: equilateral long faces exactly when $2(S - 1)^2 = (2 - S)^2 + 1$, i.e. $S^2 = 3$.

Wheels, CLEO sphere, Cleo-Icos and Antonius

- Wheel bound: n wheels with 2m spokes need $n - 1 \le m$: at most 3 (square), 4 (hexagonal), 6 (decagonal). Five octagonal: the triangles of the simple arrangement (sides 45°) have angles 65.53°, forcing pentagon angles 114.47°, against 122.25° for the equilateral equiangular spherical pentagon of side 45°.
- CLEO sphere: icosidodecahedron of edge $1/\varphi$ (circumradius 1) = 20 CLEO + 12 pentagonal pyramids (volume $\sqrt{5}/12$ each); total $(11\sqrt{5} - 5)/6$. Each pentagonal pyramid = 5 right pyramids of volume $\sqrt{5}/60$.
- Cleo-Icos: side edges 0.851, base edge $2/\sqrt{5} = 0.894$; Mackay's unit tetrahedron (edge ratio $1/\sin 72° = 1.0515$); Hypsicles' pyramids (inradius ÷ circumradius 0.79465). Dihedral angles 72° (side edges) and 69.09° (base edges).
- Prolate golden rhombohedron (edge 0.851, dihedral angles 72° and 108°), $0 \le x, y, z \le 1$ cut by $x + y + z = 1$ and 2: Cleo-Icos + Antonius + Cleo-Icos. Antonius: trigonal antiprism, two equilateral faces (edge $2/\sqrt{5}$), six side faces (0.851, 0.851, 0.894); 4 × Cleo-Icos volume; rhombohedron 6 ×. Antonius's dihedral 110.91° + 69.09° = 180°. Patch of 2,187 pieces checked. Dehn invariant of Antonius exactly −2 times the Cleo-Icos's.
- Octahedron = 20 Cleo-Icos + 12 half Admiral Caps (volumes $\varphi^2/24$, $\varphi/24$, $\varphi^6/6$ at icosahedron edge 1) = 8 Cleo-Icos + 12 kite pyramids; Cleo-Icos and half caps (with or without Antonius) cannot fill space (π, $\arccos(-\sqrt{5}/3)$, $\arccos(1/\sqrt{3})$ rationally independent).

Sources

Searched 14 October 2026: reference library, web, OEIS. Known and credited: the isogonal hexagon, antiprism angles (OEIS A394077), Odom and van de Craats 1983, Kabai 2008 and 2014, Damasceno et al. 2012, Okamoto et al. 2013, Coxeter 1973, Mackay 1962, Ino 1969, Marks 1984, Hypsicles, Hafner and Žitko, Hafner and Komatar 2008, Prodan et al. 2017, Fang et al. Scripts: studies/2026-10-14-fuji-family, -icosa-cleo-pair, -half-cap.

A.28 The DICTO Truncated-Icosahedral Hecatodihedron (DICTO DOI: 10.5281/zenodo.23360857)

Scale: the truncated icosahedron of edge $1/\varphi$ (the scale of the Dharma build).

Faces	Number	Shape	Distance from the centre
Regular pentagons	12	side $1/\varphi = 0.61803$	1.4384
Rhombi	30	sides 0.65097, angles 56.68° and 123.32°	1.5
Isosceles triangles	60	sides 0.61803, 0.65097, 0.65097	1.5

- Corners 80: the truncated icosahedron's 60 (distance 1.5315) and the 20 Shield apices (1.6057), forming a regular dodecahedron of edge $3/\varphi^2 = 1.14590$. Edges 180; two edge angles, 159.45° and 163.53°. $80 - 180 + 102 = 2$. Volume 14.4042.
- 90 faces (rhombi and triangles) at distance 1.5, the truncated icosahedron's midsphere; the 12 pentagons at 1.4384: no insphere. (Capturing the In-betweens, Version 3, corrects the earlier statement.)
- Exact, in $\mathbb{Q}(\sqrt{5})$ (data/tih_v2_check.py): the canonical pentakis dodecahedron (the truncated icosahedron's polar in its own midsphere, radius 3/2) has 12 five-fold and 20 three-fold vertices; the 20 Shield apices are its 20 three-fold vertices. The Hecatodihedron is the convex hull of the truncated icosahedron and these 20 points. Its face planes are the 90 face planes of the joined

truncated icosahedron jtI (topologically the rhombic enneacontahedron) and the 12 pentagon planes: 102 = 90 + 12. It is the jtI with its 12 five-fold corners cut off at the pentagon planes.

- Combinatorially: Conway's k6tI (80 corners, 132 faces) with 30 pairs of coplanar triangles merged into 30 rhombi (Version 2 corrected “60 pairs” to 30).
- Hecatogdoecontahedron: with its polar in the same sphere, the DICTO Ogdoecontahedron, a self-dual compound of 182 faces and 182 corners; hull the jtI, common core the rectified truncated icosahedron (see A.34a).

Sources and scripts

Searched 13–14 October 2026 (Version 2 additions 16 October 2026): Conway notation (Conway et al. 2008; Hart): k6tI 80 corners/132 faces, k5tI 72/132, ktI 92/180; Johnson solids and near misses (Johnson 1966; Zalgaller 1969; Kaplan's rectified truncated icosahedron, 92 faces); MathWorld and PolyhedronData; Polytope Wiki; rhombic enneacontahedron (jtI); the truncated icosahedron–pentakis dodecahedron compound; McCooey's Visual Polyhedra. Not found named; a report of where we looked, not a claim that the solid is unknown. Scripts: data/tih_v2_check.py, tih-v2-check.txt, tih_check.py, tih-checks.json (from Polyhedraverse's shape TRANS_DHARMA_80), tih_fig.py.

A.31 The DICTO Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688)

Scale and exactness

Coordinates in the scale of the reversed cycle (polar reciprocation in the unit sphere); all exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

The reversed cycle

Spheres of Dharma (the Ritsudō cycle): icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger. With the Hecatodihedron as seventh station, reciprocated exactly in $\mathbb{Q}(\sqrt{5})$:

dodecahedron ⊂ icosidodecahedron ⊂ Ogdoecontahedron ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ^3 larger.

Reciprocating the six solids of the cycle alone gives classical solids only. (The #30 paper calls the OG the second station, counting the inner dodecahedron as station 0.)

The solid

- 80 faces, 102 corners, 180 edges (102 − 180 + 80 = 2). 20 regular hexagons (from the Hecatodihedron's 20 Shield apexes) and 60 kites (from the truncated icosahedron's 60 corners). Corners: 12 where five faces meet, 30 where four meet, 60 where three meet.
- Edges: hexagon side squared $2/9 - 2\sqrt{5}/27$ (side 0.237881); each kite has two sides of that length, next to the hexagons, and two of squared length $86/3249 + 2\sqrt{5}/361$ (side 0.197124). 120 edges of the first length, 60 of the second.
- Angles: hexagons regular (all 120°); kite angles 117.85° (twice), 68.62° and 55.69°.
- Sphere: 90 corners exactly on the sphere of radius 2/3; the other 12 at squared radius $(170 + 2\sqrt{5})/361$. (The Hecatodihedron has 90 faces touching one sphere and its 12 pentagons a little inside it.)
- Face distances, no insphere: hexagons at squared distance $2/9 + 2\sqrt{5}/27$ (0.6228), kites at $(42 + 2\sqrt{5})/109$ (0.6529).
- No midsphere: the 120 hexagon edges touch the sphere of squared radius $7/18 + \sqrt{5}/54$ (0.65597); the 60 short kite edges come no nearer than their inner ends, on the sphere of radius 2/3.
- Dihedral angles (float): 156.20° on the 120 kite–hexagon edges, 156.72° on the 60 kite–kite edges.

- Volume $(-1000 + 1240\sqrt{5})/1539$. Convex and centrally symmetric (checked exactly); full icosahedral symmetry.
- Relation (Version 2): the canonical pentakis dodecahedron (the truncated icosahedron's polar in its midsphere, at the matching size) with its 20 three-fold corners cut off at the truncated icosahedron's hexagon planes; an unnamed solid derived from a known one (#28, Version 2).

In the reversed cycle

- Its 90 sphere corners lie exactly on the icosidodecahedron's circumsphere (radius 2/3); 30 of them, its two-fold corners, are the icosidodecahedron's own corners.
- Its 20 hexagons lie in the planes of the icosidodecahedron's 20 triangles, each hexagon holding a triangle; its 60 kites lie in the faces of the surrounding pentakis dodecahedron.

Face arrangement

Combinatorially the pentakis dodecahedron with its 20 six-fold corners cut down to the edge midpoints; in Antiprism notation (Rossiter) the dual of L0 applied to the dodecahedron (first reported in #30, §2). Checked by an exact isomorphism search on the face maps (mirror images allowed), with controls: randomly relabelled corners still match; swapping two corners of one hexagon breaks the match. L0 was built from Antiprism's written definition; the program itself was not run. McCooey's Visual Polyhedra (553 solids, scanned in full) has no solid with its counts or its dual's; nearest catalogued relative the 6-truncated pentakis dodecahedron (20 hexagons, 60 pentagons). The printed catalogues (Wenninger, Holden, Pugh) were ruled out by scope, not read.

Packings

- A copy shifted by $t = 2dn/|n|^2$ (n the hexagon's normal, d its distance) puts its opposite hexagon exactly on the shared one, with no rotation (central symmetry).
- Diamond network: joints through a tetrahedral set of four hexagons (pairwise cosine $-1/3$), $|t|^2 = 8/9 + 8\sqrt{5}/27$. Over 60 sites (1,770 pairs) no two copies overlap (exact test: two copies overlap if and only if half their offset lies inside the solid, since $K - K = 2K$); the only contacts are full-hexagon joints. Joint hexagon centres at the corners of a regular tetrahedron; joints, at the diamond bond midpoints, on the pyrochlore lattice.
- Body-centred cubic lattice packing (two networks): density exactly $(9225 - 3915\sqrt{5})/608 \approx 77.43\%$; densest packing of equal spheres 74.05% (Hales 1998).
- Local lattice search (Betke–Henk style, 200 starts, float): density 0.789260, each copy touching 12 neighbours (10 on hexagons, 2 on kites); all 200 starts converged to it; checked as a packing for lattice coefficients up to 4 (float); not proved densest.

Quasicrystal tests (canonical icosahedral quasicrystal: primitive six-dimensional lattice, triacontahedral window)

- Known case first: the triacontahedron spanned by the six unit vectors fits the window, all edges along five-fold axes.
- Lift: corners integer in six dimensions only after scaling by 171; perpendicular images reach radius 116 against a window of 1.9; first fit after nine golden inflations (solid radius about 4,750 tiling edges). Any finite golden point set fits eventually.
- Edges: none of the 180 runs along a symmetry axis (every tiling edge runs along a five-fold axis).
- Shells: the 30 two-fold corners are the icosidodecahedron's, the 30-atom shell of the Tsai and Mackay clusters (Takakura et al. 2007); the other 60 sphere corners lie 16.47° from the nearest five-fold axis (Bergman 60-shell 20.08°, Tsai 60-shell 17.78°).
- Faces: no face corner angle can be made from golden-rhombus angles (63.43°, 116.57°) or Penrose angles (multiples of 36°) with up to six tiles at the corner.
- Not tested: other six-dimensional lattices (face- or body-centred), other windows, decorated tilings. The 20 hexagon directions contain exactly five inscribed cubes, hence five body-centred cubic packings (#30).

Sources and scripts

Not found in DICTO's reference library, McCooey's catalogue and web searches (16 October 2026; printed catalogues ruled out by scope, not read): the metric Ogdoecontahedron; its exact hexagon-joined packings. Scripts (branch research of github.com/DICTOR-Master/kaleidohedra): studies/2026-10-16-polar-dharma (`polar.py`, `hstar_render.py`, `render_hstar.html`); studies/2026-10-16-flip-test (`flip.py`); studies/2026-10-16-ocd-pyrochlore (`ocd_dl0d.py`, `ocd_catalogue_scan.py`, `network.py`, `densest.py`); studies/2026-10-16-ogdoecontahedron-bridge (`bridge.py`, `bridge.txt`; commit 3fd407f). References in the paper: Firsching 2015; Betke & Henk 2000; Hales 1998; Rossiter, Antiprism; McCooey, Visual Polyhedra; Takakura et al. 2007.

A.31a The DICTO OCD Stella (DICTO DOI: 10.5281/zenodo.23430870)

Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float. OCD = the DICTO Ogdoecontahedron, in its body-centred cubic packing (cell volume $512/729 + 256\sqrt{5}/729$).

The companion C and the OCD Stella S

- C: convex hull of the 60 golden adapter pieces round one OCD (the full orbit of the 24 per cell): 72 corners, 30 hexagons in the golden planes of the OCD's envelope, 20 triangles in the three-fold planes through the adapters' outer corners (squared distance $14/27 - 10\sqrt{5}/243$); volume $(-18880 + 9760\sqrt{5})/2187$. Combinatorial type Conway t3jD (Antiprism).
- S = C cut by the packing's Voronoi cell (along the 8 joints): 48 corners; 50 faces (6 golden hexagons, 24 golden quadrilaterals, 12 triangles of C, 8 triangles in the joint hexagon planes); volume $(-4064 + 2240\sqrt{5})/729 \approx 1.296011$; pyritohedral (24 symmetries). Packing fill $151/2 - 267\sqrt{5}/8 \approx 87.12\%$, face to face across the 8 joints.

The seed

Outer S at scale φ^2, 12 copies on its five-fold axes, one at $1/\varphi$ in the centre. Of the 66 pairs, 30 have gauge exactly 1 (touching, along the icosahedron's edges) and 36 exactly φ: the seed closes exactly at φ^2 (Minkowski argument); gap $-69 + 31\sqrt{5} \approx 31.8\%$.

The crystals

Crystal	Fill (exact)	Space group (float)
OCD Stellas alone (both networks)	$151/2 - 267\sqrt{5}/8 \approx 87.12\%$	Im-3 (No. 204)
+ icosahedra (pyrochlore)	$1523/32 - 669\sqrt{5}/32 \approx 84.60\%$	Fd-3 (No. 203)
+ octahedra (pyrochlore)	$1423/32 - 3113\sqrt{5}/160 \approx 96.33\%$	Fd-3 (No. 203)
+ octahedra (rock salt, a quarter of the sites)	$3839/64 - 8453\sqrt{5}/320 \approx 91.72\%$	Fm-3 (No. 202)

- Icosahedron of the OCD's 20 hexagon planes: volume $(-80 + 80\sqrt{5})/81$. Gap pieces (cones from each centre over free faces): 12 wedges per icosahedron, $(308 - 116\sqrt{5})/2187$ (6 corners, 5 faces); per OCD Stella 12 caps, $(346204 - 143140\sqrt{5})/2381643$ (9 corners, 7 faces), 6 slabs, $(39896 - 17240\sqrt{5})/216513$ (16 corners, 12 faces), 24 thin wedges, $(12596 - 5596\sqrt{5})/88209$ (6 corners, 5 faces). Icosahedra's share of the gap $(961 + 209\sqrt{5})/2454 \approx 58.2\%$.
- Octahedron of the eight joint planes, $|x| + |y| + |z| \le R$, $R = 2\varphi/3$ (cube axes): 9/8 of the cell; each of six tips $(32 + 16\sqrt{5})/2187 = 1/48$ cell past a square wall. Notch $392/2187 - 88\sqrt{5}/1215 \approx 0.017287$ (six notches: 8.00% of the Stella); notched OCD Stella $-1616/243 + 12784\sqrt{5}/3645 \approx 1.192289$. No two octahedra may be square neighbours.
- Atom 13 (float): twelve Stellas at distance 1.56743 from the centre, gap 0.31 between them, 49.82% of the outline filled; twenty icosahedra fit up to 0.4878 of the crystal's icosahedron; 33 solids fill 58.20% of their outline.

Sources and scripts

Known: C's type (t3jD, the chamfered icosahedron); Kabai's 54-Faced Space-Filling Polyhedron (6 golden rhombi, 48 triangles, bcc, fills space); Cadotte et al. 2016; Torquato and Jiao 2009. Checked: Wikipedia (Chamfer), Klitzing, McCooey, SageMath, netlib (142 entries), Antiprism 0.29, web search for the fill and the name (17 October 2026); 64 full-text entries of DICTO's reference library (234 entries; 12 image-only and 158 link-only not text-searched). Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: studies/2026-10-16-ocd-pyrochlore (stella_cut_exact.py, stella_seed_exact.py, pyro_icosahedra.py, crystal_gap_exact.py, stella_octa_pyro.py, crystal_symmetry.py, stella_atom13.py) and studies/2026-10-20-stella-octahedra (stella_octa_exact.py).

A.34a The DICTO Hecatogdoecontahedron (DICTO DOI: 10.5281/zenodo.23430868)

Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float. H = the DICTO Hecatodihedron, O = the DICTO Ogdoecontahedron. Reciprocation in the sphere of squared radius 9/4, the midsphere of the Dharma's truncated icosahedron (touching H's 90 rhombi and triangles); the polar of H is O at its compound size. Exact: O's 90 sphere corners are the truncated icosahedron's 90 edge midpoints; no corner of H inside O and none of O inside H.

Part	Corners	Faces
H, the DICTO Hecatodihedron	80	102: 12 pentagons, 30 rhombi, 60 triangles
O, the DICTO Ogdoecontahedron	102	80: 20 regular hexagons, 60 kites
the compound	182	182; self-dual
common core H ∩ O	90	92: the rectified truncated icosahedron (12 regular pentagons, 20 regular hexagons, 60 triangles), all corners on the midsphere
hull	92	90 rhombi: the joined truncated icosahedron (rhombic enneacontahedron), the core's polar

Corner and face counts exact, volumes float (oh_compound.py). In Progenitors: the golden cycle I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ^3.

Sources and scripts

Checked 17 October 2026: MathWorld (the truncated icosahedron–pentakis dodecahedron compound); Wikipedia (rectified truncated icosahedron; truncated icosahedron); McCooey; Hart's Virtual Polyhedra; web search for the name (no results); SageMath; netlib; Antiprism 0.29; Progenitors §4 and SEARCH-LOG.md; DICTO's reference library. Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: studies/2026-10-16-progenitors (oh_compound.py → oh-compound.txt, core_hull_in_cycle.py).

A.35a The DICTO Metaxyhexecontahedron (DICTO DOI: 10.5281/zenodo.23430864)

Exact in $\mathbb{Q}(\sqrt{5})$. Icosahedron (0, ±1, ±φ) and cyclic shifts (edge 2); dodecahedron (±1, ±1, ±1) and cyclic shifts of (0, ±φ, ±1/φ), corners over the icosahedron's face centres (edge 2/φ), scaled by k; hull edge ratio (icosahedron : dodecahedron) φ/k. M: k = $\varphi^5/9$ = 11/18 + 5√5/18, ratio $9/\varphi^4$ ≈ 1.31308 (squared $81/\varphi^8$). Closed form: hull of (0, ±1, ±φ) and ($\varphi^5/9$)·{(±1, ±1, ±1), (0, ±φ, ±1/φ)} (cyclic).

Property	Value (exact)
corners	32: 12 five-fold (the icosahedron's), 20 three-fold (the dodecahedron's)
faces, edges	60 congruent isosceles triangles, 90 edges; the pentakis dodecahedron's combinatorics
squared edge lengths	94/81 + 14√5/27 (60 edges, five-fold to three-fold); 14/9 + 2√5/27 (30 edges, three-fold to three-fold)
squared corner radii	5/2 + √5/2 (five-fold); 41/18 + 55√5/54 (three-fold)
insphere	all 60 faces at squared distance 7249/4162 + 16209√5/20810
midsphere	none: edges at squared distances 2 + 20√5/27 and 161/81 + 8√5/9
volume	410/27 + 550√5/81 ≈ 30.3684

In the cycle

- The triacontahedron reaches only 0.812 of the step RT → I·φ³; no convex icosahedral solid touches both neighbours (Vajrayana, Table 3, float).
- In the cycle's units: hull of RT's 12 five-fold corners and the 20 face centres of I·φ³ (squared radii 45/8 − 9√5/8 and 47/24 + 7√5/8); same solid, checked exactly.
- RT → M: 30 pieces of one kind; M → I·φ³: 60 pieces of one kind (triangular-bipyramid shape); M takes 0.356 of the step's volume.
- Polar: 12 regular pentagons and 20 hexagons with sides alternating in the ratio (14 + 9√5)/11.
- Family: fully symmetric for edge ratios (5 + 3√5)/10 to φ; Catalan pentakis dodecahedron 3(7 + 5√5)/38 ≈ 1.435; Palintriacontahedron 3/√5 ≈ 1.342; Metaxyhexecontahedron 9/φ⁴ ≈ 1.313. 9/φ⁴ is also the exact gain per round of the tightest progenitor cycle (noted without explanation).

Sources and scripts

Checked 17 October 2026: Polytope Wiki, Wikipedia and MathWorld (Pentakis Dodecahedron; Dodecahedron–Small Triambic Icosahedron Compound), McCooey, OEIS (9/φ⁴ matches no sequence), Bridges 2017, Wassell and Benito 2012, arXiv (closest arXiv:2008.00862), web search for the name, SageMath, netlib, Antiprism 0.29, DICTO's reference library. Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: scripts/hull_family_exact.py → data-hull-family-exact.txt (every face plane and corner tested exactly); studies/2026-10-16-progenitors (gap_member.py, gap_pieces_exact.py, metaxy_dual.py).

A.35b The DICTO Palintriacontahedron (DICTO DOI: 10.5281/zenodo.23430866)

Exact in ℚ(√5) unless marked float. Same setting as A.35a; X: k = φ√5/3 = 5/6 + √5/6, edge ratio 3/√5 (squared 9/5). Closed form: hull of (0, ±1, ±φ) and ((5 + √5)/6)·{(±1, ±1, ±1), (0, ±φ, ±1/φ)} (cyclic).

Property	Value (exact)
corners	32: 12 five-fold, 20 three-fold, all tips of the full-height icosidodecahedron star
faces, edges	60 triangles, each half a rhombic-dodecahedron rhombus (cos² of the face angles 1/9, 1/3, 1/3); 90 edges
squared edge lengths	20/9 (60 edges) and 5/3 (30 edges)
squared corner radii	5/2 + √5/2 (five-fold); 5/2 + 5√5/6 (three-fold)
insphere	all 60 faces at squared distance 7/4 + 3√5/4
midsphere	none (edges at squared distances 25/12 + 2√5/3 and 35/18 + 5√5/6)
volume	50/3 + 50√5/9 ≈ 29.0893

- Float (as labelled in Vajrayana): X lies inside the next turn's D′; 99.46% of its volume (20,000 sample points) lies inside the next icosahedron's Cleo star. The first stellation of the icosahedron

puts its tips at 3/√5 times the face distance.

- Family: Catalan pentakis dodecahedron 3(7 + 5√5)/38 ≈ 1.435; X 3/√5 ≈ 1.342; Metaxyhexecontahedron 9/φ⁴ ≈ 1.313.

Sources and scripts

Known: the hull family and its fully symmetric range (Polytope Wiki), the rhombic triacontahedron, the icosidodecahedron's central pyramids. Checked as for A.35a (Vajrayana §5, SEARCH-LOG.md; web and libraries, 17 October 2026). Script: scripts/hull_family_exact.py (exact).

A.41 The DICTO Jewel (DICTO DOI: 10.5281/zenodo.23496353)

Exact in ℚ(√5) except where marked float. Scale: the Hexa's cube has edge 1, so a cell has edge 1/2 and volume 1/8 (the scripts use cells of edge 2, every volume 64 times larger). Cells even or odd by index-sum parity; even cells form a face-centred cubic lattice.

The Jewel

- Euclid's roof: ridge 1/φ times the cell edge, height (φ − 1)/2 times the cell edge; roof (1 + √5)/192; dodecahedron (5 + √5)/32. The neighbouring stella (volume 1/16) takes (√5 − 1)/192, leaving a carved roof of exactly 1/96. Jewel = 1/8 + 6/96 = 3/16.
- Each carved roof is four tetrahedra (one per quarter of the cube face). 38 corners (cube 8, ridge points 12, face centres 6, window-completing points 12); 60 faces (12 windows, 48 triangles in the stellas' planes); 38 − 96 + 60 = 2.
- Window: thick Penrose rhombus, side (√5 − 1)/4 = 1/(2φ), long diagonal 1/2, 108° at the ridge corners (cos 108° = (1 − √5)/4) and 72° at the cube corners.
- Symmetry: order 24, pyritohedral (centrosymmetric, three-fold turns, three coordinate mirrors; no diagonal mirror, no four-fold turn, no (111) mirror).

The Stella–Jewel Lattice

Odd cell = stella + six carved roofs: 1/16 + 6/96 = 1/8 (the uncarved roofs meet in volume exactly 0). Jewels and stellas tile space, 3/16 : 1/16 = 3 : 1.

Clusters

piece	faces	volume	Jewels	stella inside	loose stellas per piece	lattices	up to symmetry
Jewel-Tetra	228 (192 triangles, 36 windows)	3/4	4	0	4	4 (index 8)	2
Jewel-Octa	288 (240, 48)	19/16	6	1	5	12 (index 12)	1
Jewel-Rhombi	396 (336, 60)	25/16	8	1	7	25 (index 16)	9

- Neighbouring Jewels (one face-diagonal step) share exactly one window; two cells apart along an axis, none. Jewel-Rhombi: 18 windows (12 edges, 6 short face diagonals); 8⅓ Jewels.
- Jewel-Octa example lattice: Hermite basis (1,0,3), (0,1,5), (0,0,12) in cell units. Octet pattern: each Jewel in 8 Jewel-Tetras and 6 Jewel-Octas; without overlap at most one stella in six inside a Jewel-Octa, reached by the 12 lattices.
- Jewel-Tetras sharing single Jewels corner to corner form a pyrochlore network (DICTO's Kagome network).

Hexa and Hexa-Key

Hexa = block (edge 1) + 24 carved roofs = 1 + 24/96 = 5/4 (6⅔ Jewels). Hexa-Key = 1 − 24/96 = 3/4 = 8/16 + 12 × 2/96 (eight stellas, twelve double roofs). Hexas on alternate blocks (rock salt); 5 : 3. Each Hexa meets each of 12 neighbours on exactly two whole windows (24 in all); no two Hexas' roofs overlap.

Twin

Twin rule: a centrosymmetric solid with a three-fold axis has as mirror image the same solid turned 60°. The packing: a (111) twin left about 4% of points uncovered and 4% covered twice (float, #37a); the (111) mirror takes the cell step (1, 0, 0) to (1/3, −2/3, −2/3), off the cubic lattice.

Checks and sources

jewel_exact.py (21 PASS) and cluster_exact.py (24 PASS), exact; lalvani_edges.py (float): all 48 window edges lie on the icosahedral two-fold axes, none of the other 48. Face counts counted from exact shared windows and stellas. Searched: full text of 80 library sources and web, 20 October 2026. References: Euclid XIII.17 (Heath 1908); Kepler 1619; Penrose 1979; Bragg 1913; Lalvani US 4,723,382 (1988); DICTO #8, #17, #18, #37, #37a, #40, Kaleidohedra findings #10, #10b, #15, #16. Study: studies/2026-10-20-dicto-jewel, commit d475b59.

A.45 The Reuleaux Platonic Solids (DICTO DOI: 10.5281/zenodo.23529499)

For a Platonic solid S with corner set X and diameter D, $B = \cap_{v \in X} B(v, D)$, a ball polyhedron. Tetrahedron: D is the edge (the Reuleaux tetrahedron). Cube, octahedron, dodecahedron, icosahedron: D = 2R. Lengths at D = 1.

Main theorem

(i) Every width of B is at least 1, and the width along the line through any two corners at distance 1 is exactly 1. (ii) For the cube, octahedron, dodecahedron and icosahedron, with circumradius ½ and inradius r, the width along the axis through a face centre is exactly $2\sqrt{(3/4 + r^2)} - 2r$.

Proof sketch. (i) By Pál's theorem X lies in a body K of constant width 1; K ⊂ B. For corners v, w at distance 1 and u = v − w, the extremes of u·x on B are u·v and u·w. (ii) The rotations about the axis fix u, so the largest u·x is reached at a point zu on the axis; $|zu - v|^2 = ¼ + z^2 - 2z\,h_v$, largest where $h_v = -r$ (the opposite face); zu ∈ B exactly when $z \le \sqrt{(3/4 + r^2)} - r$; central symmetry doubles it.

Corollaries

- Dual pairs: $(r/R)^2 = 1/3$ (cube, octahedron) and $(5 + 2\sqrt{5})/15$ (dodecahedron, icosahedron). Cube and octahedron: $r^2 = 1/12$, $w = (\sqrt{30} - \sqrt{3})/3 \approx 1.248392$. Dodecahedron and icosahedron: $r^2 = (5 + 2\sqrt{5})/60$, $w = (\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989$.
- Tetrahedron: (ii) does not apply; along the edge-midpoint axis the width is $2\sqrt{(3/4)} - 1/\sqrt{2} = \sqrt{3} - \sqrt{2}/2 \approx 1.024944$ (exact).
- Not Reuleaux polyhedra: diametric pairs 4, 3, 10, 6 against the Vázsonyi bound 14, 10, 38, 22; each corner lies on exactly one other sphere, so it is not a vertex of B.
- Conjecture 1: the face-axis width (tetrahedron: $\sqrt{3} - \sqrt{2}/2$) is the largest. Numerical: 3000 random directions with local refinement give 1.248391589161, 1.110988655456 and 1.024944026382.

Body (DICTO's name)	Balls on	Faces	Arcs	Diametric pairs (2m − 2)	Fewest colours	Width range	Volume ≈
Reuleaux tetrahedron	tetrahedron	4	6	6 (6)	4	1 .. 1.024944	0.42216
Reuleaux octahedron	cube	8	12	4 (14)	2	1 .. 1.248392	0.64940
Reuleaux cube	octahedron	6	12	3 (10)	3	1 .. 1.248392	0.68764
Reuleaux icosahedron	dodecahedron	20	30	10 (38)	3	1 .. 1.110989	0.57188
Reuleaux dodecahedron	icosahedron	12	30	6 (22)	4	1 .. 1.110989	0.59679

At diameter 1. Arcs numerical; diametric pairs exact; width minimum proved, maximum found numerically, proved attained and conjectured largest; volumes by radial quadrature. Reuleaux tetrahedron: Harbourne's $8\pi/3 + \sqrt{2}/4 - (27/4)\arccos(1/3)$ (as quoted by Hynd) and our 0.4221577 agree to 1.1×10^{-8}.

Checks and sources

reuleaux_checks.py (45 PASS): Part A exact in $\mathbb{Q}(\sqrt{5})$ (diameters, diametric pairs, corners on one other sphere, corner heights, $(r/R)^2$, closed forms, tetrahedron axis); Part B float64 (widths, volumes by Gauss–Legendre, N = 1000 and 2000 within 6×10^{-7}; arcs with 7200 points per circle; colours by backtracking). widths.py; reuleaux_platonic.py. Known: Bezdek, Lángi, Naszódi and Papez 2007; Montejano and Roldán-Pensado 2017; Hynd 2023; Sallee 1970 (not read). Searched: 81 library sources with a text layer, 21 October 2026 (175 link-only entries not searched). Study commit f193a56.

A.49 The Polyhedral Stretching Theorem (DICTO DOI: 10.5281/zenodo.23566671)

Theorem

Let P be a convex polyhedron whose edges all have length 1, slid a distance 1 along a direction u. The swept shape (the convex hull of P and P + u) has every edge of length 1 unless u is parallel to an edge of P. Faces: each face of P not parallel to u, kept whole (at the end it faces); each n-sided face parallel to u, stretched into an (n + 2)-gon, with one side fewer for each of its edges parallel to u; one parallelogram for every edge of P's outline seen along u (a rhombus at slide 1). For the regular dodecahedron a pentagon along the slide becomes a hexagon if u runs along one of its edges and a heptagon otherwise; with no pentagon along the slide the belt has 10 or 12 rhombi (in 2 000 random directions; the count is observed, the equal edges proved).

Proof sketch. The swept shape is the Minkowski sum $P \oplus S$, S the segment from 0 to u. Every face of a Minkowski sum is the sum of a face of each summand in a common outward direction n: S contributes u ($n \cdot u > 0$), 0 ($n \cdot u < 0$) or S itself ($n \cdot u = 0$). Edges are translates of edges of P (length 1) or of S (length 1), except where an edge of P is parallel to u: there $e \oplus S$ is one segment of length 2. Corollary (caps): the same with S replaced by the triangle or tetrahedron of side 1.

Slide along	Faces	Every edge 1?
a cube axis (DICTO's F11)	8 pentagons, 4 hexagons, 2 rectangles (squares at slide 1)	no: the slide runs along two edges
a face diagonal of Euclid's cube	12 pentagons, 10 rhombi	yes
a body diagonal (three-fold axis)	12 pentagons, 12 rhombi (six of them squares)	yes
a five-fold axis	12 pentagons, 10 rhombi, all alike (angles 58° and 122°)	yes
along two pentagons (example)	10 pentagons, 6 rhombi, 2 heptagons	yes

- Face-diagonal slide at length 1: rhombi of three shapes (2 with 45°/135°, 4 with 55°/125°, 4 with 69°/111°); telescoped to $\varphi\sqrt{2} \approx 2.288$, to the next dodecahedron.
- Three caps: 12 pentagons, 15 rhombi, 2 equilateral triangles (29 faces); four caps: 12 pentagons, 18 rhombi, 4 equilateral triangles (34 faces).
- Turned caps: no triangle belt; worst edge off by 0.336 at best.
- Other Platonic solids (1 000 random stretches each): tetrahedron with four caps (opposite tetrahedron) = cuboctahedron; octahedron with four caps = truncated tetrahedron; tetrahedron through a corner = elongated triangular pyramid (J7); three caps on a triangle turned against its base = triangular cupola (J3); octahedron through a corner = elongated square bipyramid (J15); the stretched cube is a box, a hexagonal prism or a zonohedron.

Icosahedron	Faces
five-fold axis	20 triangles, 10 rhombi
face diagonal	16 triangles, 2 rhombi, 4 pentagons
body diagonal	20 triangles, 6 rhombi
cube axis	16 triangles, 4 trapezoids, 2 squares; not equal-edged (two edges become 2 long)
three caps	15 triangles, 3 rhombi, 3 pentagons, 2 hexagons
four caps	16 triangles, 6 rhombi, 4 hexagons

Checks and sources

void_volume.py (the void between two dodecahedra on edge-sharing cubes: volume $\varphi^3 + \frac{1}{2}$, no overlap); sweep_theorem.py; caps34.py; gyro_caps.py; ring1.py and identify.py; icosa_family.py; nets.py and telescope.py (unfoldings checked flat with no overlaps). Telescoping model extends to about 2.7 edges. Known: Minkowski sums (Deza and Pournin 2018; Erdahl and Grishukhin 2015); Hart and Picciotto; Arseneva, Langerman and Zolotov 2020; Baumann's near misses; Grünbaum 2010 (after Fedorov). Searched: reference library (Appendix C of the paper) and web, 22 October 2026. Study commit 014b5ee.

A.51 The Delian Bond (DICTO DOI: 10.5281/zenodo.23606828)

Setting

Octet of edge 1: regular tetrahedra, volume $V_T = \sqrt{2}/12$, and regular octahedra, $V_O = \sqrt{2}/3 = 4V_T$, two tetrahedra per octahedron. Each tetrahedron is 4 cones from its centre c_T, each octahedron 8 cones from its centre c_O (the face pyramids, which fill space together, #44). For a shared face F the double cone B_F runs from c_T through F to c_O; c_Tc_O is perpendicular to F and meets it at its centre; F lies one third of the way from c_T to c_O (inradii $1/(2\sqrt{6})$ and $1/\sqrt{6}$).

Classical facts used: the rhombic dodecahedron is an octahedron with the 8 tetrahedron cones over its faces added (its 8 three-fold corners at neighbouring tetrahedron centres; volume $V_O + 8 \cdot V_T/4 =$

$6V_T$), and equally a cube with six square pyramids; the cube of edge $1/\sqrt{2}$ is a tetrahedron with the 4 octahedron cones over its faces added (other four corners at neighbouring octahedron centres; volume $V_T + 4 \cdot V_O/8 = 3V_T$).

Main theorem

For $0 \le t \le 1$ cut every B_F by the plane parallel to F through $c_T + t(c_O - c_T)$; T(t) = the 4 parts next to a tetrahedron centre, O(t) = the 8 parts next to an octahedron centre. They fill space face to face and both are convex.

- (a) $0 < t \le 1/3$: T(t) regular tetrahedron of edge 3t, volume $27t^3 \cdot V_T$; O(t) rhombic dodecahedron with its 8 three-fold corners cut by planes parallel to the octahedron's faces (8 triangles and 12 hexagons for $t < 1/3$), volume $(6 - 54t^3) \cdot V_T$.
- (b) $1/3 \le t < 1$, $a = 3(1 - t)/2$: O(t) regular octahedron of edge a, volume $4a^3 \cdot V_T$; T(t) cube with its 4 corners at octahedron centres cut (4 triangles and 6 hexagons for $t > 1/3$), volume $(3 - 2a^3) \cdot V_T$.
- (c) $t = 0$: rhombic dodecahedra alone; $t = 1/3$: the octet; $t = 1$: cubes alone.

Proof sketch. The double cones fill space and each is divided by one plane, so the pieces fill space. For $t \le 1/3$ the plane is in the tetrahedron's cone at fraction 3t from c_T to F: the parts next to c_T make the tetrahedron scaled by 3t; the parts next to c_O are the rhombic dodecahedron cut by 8 planes (c_T is its only corner beyond the plane of F), convex as an intersection of convex sets; each plane cuts the three edges at a three-fold corner at fraction 3t. For $t \ge 1/3$ the plane is at fraction $(3/2)(1 - t) = a$ from c_O to F, and the roles exchange (octahedron scaled by a; cube cut by 4 planes). Volumes: (a) $27t^3V_T$ and $6V_T - 8 \cdot (V_T/4) \cdot 27t^3$; (b) $a^3V_O = 4a^3V_T$ and $3V_T - 4 \cdot (V_O/8) \cdot a^3$. Face to face: each triangle is the cut face of its neighbour of the other kind; hexagons are shared by neighbours of the same kind, cut by the same planes.

Check. Exact, in rational arithmetic, at t = 1/10, 1/5, 1/3, 1/2, 4/5, 9/10 (octet/octet_exact.py → octet-exact.txt).

Theorem 2 (special members)

- (a) $t = (2\sqrt{6} - 3)/15$: every edge (tetrahedra and cut rhombic dodecahedra) has length $(2\sqrt{6} - 3)/5$. From $3t = (\sqrt{6}/4)(1 - 3t)$, the rhombic dodecahedron edge being $\sqrt{6}/4$.
- (b) $t = 5/3 - (2/3)\sqrt{2}$: every edge (cut cubes and octahedra) has length $\sqrt{2} - 1$. From $a = (1 - a)/\sqrt{2}$, the cube edge being $1/\sqrt{2}$.
- (c) $t = 1 - \sqrt[3]{6}/3$: tetrahedron pieces and octahedron pieces each fill exactly half of space. From $2(3 - 2a^3) = 4a^3$, $a^3 = 3/4$ (for $t \le 1/3$ the tetrahedron pieces take at most a third).
- (d) For no t is a hexagon regular: cut-cube hexagons have angles 90°, 135°, 135°, 90°, 135°, 135°; cut-rhombic-dodecahedron hexagons keep acute angles of about 70.5° (144.7° at the cut corners).

Check. (a)–(c) exactly in $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt[3]{6})$, with no floating point in any equality (octet/octet_special_exact.py → octet-special-exact.txt).

Theorem 3 (the Delian bond)

T(t) and O(t) have equal volume at exactly one position, $t_D = 1 - (2/3) \cdot 2^{-1/3}$. There Samson (the cut cube $T(t_D)$) and Delilah (the octahedron $O(t_D)$) both have volume $\sqrt{2}/6 = 2V_T = V_O/2$. Delilah is the octet's octahedron shrunk by $2^{-1/3}$: edge $1/\sqrt[3]{2}$. Samson's edges: $1/\sqrt[3]{2}$ (his triangles, Delilah's faces) and $(1 - 2^{-1/3})/\sqrt{2}$.

Proof sketch. For $t \le 1/3$, vol T(t) $\le V_T$ and vol O(t) $\ge 4V_T$: never equal. For $t \ge 1/3$, $(3 - 2a^3)V_T$ decreases and $4a^3V_T$ increases with a, so at most one solution: $3 - 2a^3 = 4a^3$, $a^3 = 1/2$, $a = 2^{-1/3}$, $t_D =$

$1 - 2a/3$. **Check:** exact in $\mathbb{Q}(\sqrt[3]{2})$: equal volumes, Delilah's edges, Samson's two edge lengths.

Table: the slider (octet edge 1, $V_T = \sqrt{2}/12$)

Position	t	Tetrahedron piece	Octahedron piece
rhombic dodecahedra	0	—	rhombic dodecahedron, $6V_T$
equilateral (a)	$(2\sqrt{6} - 3)/15 \approx 0.1266$	tetrahedron, edge $(2\sqrt{6} - 3)/5$	cut rhombic dodecahedron, same edge
the octet	1/3	tetrahedron, edge 1, V_T	octahedron, edge 1, $4V_T$
half and half	$1 - \sqrt[3]{6}/3 \approx 0.3943$	cut cube, $3V_T/2$	octahedron, $a = \sqrt[3]{6}/2$, $3V_T$
Delian bond	$1 - (2/3)\cdot 2^{-1/3} \approx 0.4709$	**Samson**, $2V_T$	**Delilah**, edge $2^{-1/3}$, $2V_T$
equilateral (b)	$5/3 - (2/3)\sqrt{2} \approx 0.7239$	cut cube, edge $\sqrt{2} - 1$	octahedron, edge $\sqrt{2} - 1$
cubes	1	cube, edge $1/\sqrt{2}$, $3V_T$	—

The three DICTONIC BONDS (names by DICTO)

Bond	Solids	From	Volumes
Golden	Icosaxes + Unborn Axes	icosahedra, Unborns and AXE, cut at the golden point (#48)	$1 + 2\sqrt{5}/3$ and $1 + \sqrt{5}/3$
Keyed	Lotus Jewel Axes + Dodecaxes	the Lotus Jewel network, cut the same way (#48)	$\sqrt{5}$ and 2
Delian	Samson + Delilah	the octet, cut where the volumes are equal (#51)	$\sqrt{2}/6$ each

Golden and keyed volumes at icosahedron edge 1, Delian at octet edge 1. The Icosaxes and Unborn Axes are the truncated octahedra of Koca et al. ($x = \tau$ and $x = 1/\tau$), reached by a new route; the Lotus Jewel Axes and Dodecaxes are new (#48).

Remark: the cube

The same weighted cut between a lattice of cubes and the centres of its cubes, with weight w (a power diagram): regular truncated cubes with octahedra at $w = 1/4 - 1/\sqrt{2}$ (every edge $\sqrt{2} - 1$ at cube edge 1); cuboctahedra with octahedra at $w = -1/4$ (#48's $x \to 0$); Kelvin's truncated octahedra at $w = 0$ ($x = 1$); in between, #48's T(x) and T(1/x) with $w = (x - 1)/(4(x + 1))$. The equal-edged members are the uniform truncated cubic and rectified cubic honeycombs; volumes are equal only for Kelvin's cell. The ratio 2 : 1 occurs at $x \approx 0.3747$, a root of $16u^3 - 36u^2 + 18u - 1 = 0$, irreducible with three real roots, so with no expression in real radicals. No new bond; checked exactly (changeover/changeover_exact.py → changeover-exact.txt).

Remark: the golden bond, polar

DICTO's idea: the solids turned inside out (polar reciprocation about their centres). The polar of #48's T(x) (squares at distance $(1 + x)s$, hexagons at $(1 + 2x)s/\sqrt{3}$, scale s) is a cube with a low pyramid on each face (14 corners, 24 triangles, the type of the tetrakis hexahedron), pyramid height $x/(1 + x)$ in its own half cube edges: $1/\varphi$ for the Icosaxes ($x = \tau$), $1/\varphi^2$ for the Unborn Axes ($x = 1/\tau$), 1/2 for Kelvin's cell; for each pair x, 1/x the heights add to 1, the height of the pyramids that make a cube into the rhombic dodecahedron. A statement about proportions (the two polars have cubes of different sizes). Polar of Delilah: a cube; of Samson: an octahedron with low pyramids on four alternate faces; the keyed solids are not convex and have no polar solid. The polar solids are not claimed to fill space.

Sources and scripts

Known: the octet (tetrahedral-octahedral honeycomb); the rhombic dodecahedron as a cube with six pyramids and as an octahedron with eight quarter-tetrahedra; the cube as a tetrahedron with four

corners; Knoll and Morgan 2003 (cube packing, octet and rhombic dodecahedron packing related as separate steps); power (Laguerre) diagrams (Aurenhammer 1987), whose power bisectors of the tetrahedron and octahedron centres are the cut planes here; Conway, Jiao and Torquato 2011; Gabbrielli, Jiao and Torquato 2012; Kelvin 1887; the uniform truncated cubic and rectified cubic honeycombs. Checked: DICTO's reference library (nothing on this family; Gabbrielli–Jiao–Torquato 2012 read for this check); Knoll and Morgan 2003 read in full; web searches, 23 October 2026 (cubes with four alternate corners truncated with octahedra; rhombic dodecahedra truncated at their three-fold corners with tetrahedra; power diagrams of the tetrahedral and octahedral holes of the face-centred cubic lattice; tetrahedra and octahedra of equal volume, cube root of 2), which found only the classical octet and its relatives. Not checked: Wolfram PolyhedronData, Stella / Great Stella libraries. Not found in those sources: the family as such, its two equilateral members, the half-and-half member, the Delian bond, and the golden division of the rhombic dodecahedron's pyramid by the polar golden bond. Scripts: study 2026-10-23-shadow-forms, commit 4e98aa8, github.com/DICTOR-Master/kaleidohedra (research branch). Doubling the cube: Heath 1921; ruler and compass impossibility: Wantzel 1837.